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## A Brief Overview of Future Mobility and Energy Management Strategies with Vehicle-to-Everything (V2X) Concept on Smart Grid

**Item Type** Conference Paper

**Author** I. Colak

**Author** E. Irmak

**Author** İ. F. Tepe

**Abstract** In recent years, there has been a notable increase in interest in electric vehicles (EVs) due to several factors, including the depletion of fossil fuels, environmental concerns such as carbon dioxide (CO<sub>2</sub>) emissions, and the fact that the vehicle industry is the main driver of CO<sub>2</sub> emissions. In line with this growing interest, there are both opportunities and challenges in the integration of EV charging systems into the electricity grid. Grid operators may begin to regard electric vehicles (EVs) as a novel distributed resource, given that the stored energy in their batteries can be employed for a multitude of applications. The concept of vehicle-to-everything (V2X), enables the batteries of electric vehicles (EVs) to participate in energy management as a distributed resource. EVs can support the grid through topologies such as vehicle-to-grid (V2G), vehicle-to-home (V2H), vehicle-to-load (V2L), vehicle-to-building (V2B), and vehicle-to-vehicle (V2V). This paper provides a summary of the benefits and challenges associated with the implementation of each V2X approach, as well as proposing potential solutions to address these challenges.

**Date** 9-13 Nov. 2024

**DOI** 10.1109/ICRERA62673.2024.10815234

**Extra** Journal Abbreviation: 2024 13th International Conference on Renewable Energy Research and Applications (ICRERA)

**ISBN** 2572-6013

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**Proceedings Title** 2024 13th International Conference on Renewable Energy Research and Applications (ICRERA)

**Conference Name** 2024 13th International Conference on Renewable Energy Research and Applications (ICRERA)

**Date Added** 20/09/2026, 7:46:33 pm

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### Tags:

Security, Vehicles, Vehicular ad hoc networks, Vehicle-to-grid, Costs, Vehicle-to-everything, Topology, Batteries, Carbon emissions, Electric potential, electric vehicle (EV), vehicle-to-everything (V2X), vehicle-to-grid (V2G)

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## A Context Aware and Traffic Adaptive Privacy Scheme in VANETs

**Item Type** Conference Paper

**Author** I. Saini

**Author** S. Saad

**Author** A. Jaekel

**Abstract** Preserving privacy in VANETs is a significant challenge for users and public acceptance of VANETs. The use of a pseudonym is a common technique for enhancing the user's privacy in VANETs. Several Pseudonym Changing Schemes (PCS) for user's privacy in VANETs have been proposed. The highly dynamic topology of the vehicular network can impact the way the pseudonymous identifiers are changed. To make these changes inconspicuous, we introduce the Context-Aware and Traffic Adaptive privacy scheme, which takes into account the rapidly changing traffic condition. In this paper, we propose a new PCS that aims to benefit the most from the context of the vehicle and traffic patterns to leverage a suitable situation for changing pseudonyms that increases anonymity. The vehicles change the pseudonym simultaneously in a region to increase privacy by maximizing the anonymity set. The proposed approach is evaluated in the presence of an adversary actor who could engineer privacy attacks against any given PCS.

**Date** 18 Nov.-16 Dec. 2020

**DOI** 10.1109/CAVS51000.2020.9334559

**Extra** Journal Abbreviation: 2020 IEEE 3rd Connected and Automated Vehicles Symposium (CAVS)

**Pages** 1-5

**Proceedings Title** 2020 IEEE 3rd Connected and Automated Vehicles Symposium (CAVS)

**Conference Name** 2020 IEEE 3rd Connected and Automated Vehicles Symposium (CAVS)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Privacy, Traffic control, Network topology, Vehicle dynamics, Context-aware services, Road transportation, Topology

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## A Hybrid Model to Investigate Perception Towards Adoption of Autonomous Vehicles in India

**Item Type** Conference Paper

**Author** A. Bhardwaj

**Author** Y. Kumar

**Author** N. Hasteer

**Abstract** Autonomous Vehicles are becoming new trend in automobile industry as more and more manufacturers are trying to develop self-driving vehicles. This decade will mark the shift from people driving cars to cars driving people to their destination. As many level 4 and level 5 autonomous vehicles are under development and many level 2 and level 3 are available in market, it is pertinent to investigate what and how much people know and understand about this new technology and what is their perception towards using autonomous vehicles in India. The aim of this study is to gain an understanding of consumer perspective towards autonomous vehicles in India. A Hybrid model based on unified theory of acceptance and use of technology (UTAUT) and technology acceptance model

TAM was used to assess people's comfort with levels of autonomy, acceptance of autonomous vehicles and other related attributes about perception. The model was implemented and public perception was collated through an online survey onto which statistical analysis was performed. From the results, we infer that people are excited to use level 5 vehicles. Though Indian public is waiting for these vehicles to arrive in market yet there appear some apprehensions about security.

**Date** 24-26 Sept. 2021

**DOI** 10.1109/GUCON50781.2021.9573588

**Extra** Journal Abbreviation: 2021 IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)

**Pages** 1-5

**Proceedings Title** 2021 IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)

**Conference Name** 2021 IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

autonomous vehicles, Security, Automobiles, Transportation, Technological innovation, Technology acceptance model, self-driving, Market research, consumer perceptions, Statistical analysis

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## A LoRa-Powered Secure Vehicular Communication System: Internet of Vehicles (IoV)

**Item Type** Conference Paper

**Author** T. Sharma

**Author** R. Daruwala

**Abstract** The implications of the Internet of Vehicles (IoV) concept grow with the number and variety of connected vehicles. Safe transportation becomes important when vehicle users seek comfort on the road. Vehicle-to-everything (V2X) communication technology enhances communication between vehicles and infrastructure. To meet the growing demand for connected vehicles, network speed, energy efficiency, and spectrum consumption must be improved over ad-hoc or cellular connectivity. The current research utilizes Low-Power, Wide-Area Networking (LPWAN) technology as Long-Range (LoRa). Scalability, flexibility, long-range, and low power consumption are all provided by LoRa communication over the Long Range-Wide Area Network (LoRaWAN). The current work is a proof-of-concept for a hardware prototype on a Vehicular Ad-Hoc Network (VANET) that employs LoRa modules and a LoRaWAN gateway. The proposed system is tested for Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication between LoRa devices, and it sends data to a LoRaWAN server via the gateway. The test results significantly support the usefulness of the proposed method for establishing a secure connection in connected vehicles. The acquired metrics of the better-received Signal Strength Indicator (RSSI) and Signal Noise Ratio (SNR), support the proposed method for enhanced communication utilizing LoRaWAN for connected vehicles.

**Date** 14-15 May 2024  
**DOI** 10.1109/WFPST58552.2024.00005  
**Extra** Journal Abbreviation: 2024 IEEE World Forum on Public Safety Technology (WFPST)  
**Pages** 149-152  
**Proceedings Title** 2024 IEEE World Forum on Public Safety Technology (WFPST)  
**Conference Name** 2024 IEEE World Forum on Public Safety Technology (WFPST)  
**Date Added** 20/09/2026, 7:46:33 pm  
**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

Vehicular ad hoc networks, User experience, security, Communication, Connected vehicles, VANET, protocol, Measurement, Logic gates, LoRaWAN, Vehicle-to-infrastructure

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## A Proposed Safety Case Framework for Automated Vehicle Safety Evaluation

**Item Type** Conference Paper  
**Author** J. Wishart  
**Author** J. Zhao  
**Author** B. Woodard  
**Author** G. O'Malley  
**Author** H. Guo  
**Author** S. Rahimi  
**Author** S. Swaminathan  
**Abstract** Automated driving system (ADS)-equipped vehicles (AVs) are currently being developed and deployed on public roads. While the benefits to public safety (among other potential benefits) are promising, the risk that AVs pose to public safety is not yet understood. The Automated Vehicle Test and Evaluation Process (AV - TEP) mission initiated by Science Foundation Arizona in collaboration with Arizona State University is intended to provide a framework for an AV developer or third-party evaluator (such as a regulator) to use in order to provide evidence that the AV is safe for its intended implementation. The AV-TEP framework uses the safety case concept that is widely accepted in the AV industry and others, and consists of three pillars: (1) Safety Management System (SMS), (2) Design Methods, and (3) Scenario-Based Testing. The three pillars are described in detail, and the validation methodology used is outlined. Finally, the various paths to industry acceptance of the AV-TEP framework are described, which builds upon existing work and research wherever appropriate.  
**Date** 16-18 Oct. 2023  
**DOI** 10.1109/IAVVC57316.2023.10328077  
**Extra** Journal Abbreviation: 2023 IEEE International Automated Vehicle Validation Conference (IAVVC)  
**Pages** 1-7  
**Proceedings Title** 2023 IEEE International Automated Vehicle Validation Conference (IAVVC)  
**Conference Name** 2023 IEEE International Automated Vehicle Validation Conference (IAVVC)  
**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

digital twin, scenario-based testing, Safety management, Collaboration, Roads, Public security, Industries, Automated Driving System (ADS), Design methodology, Driving Safety Assessment (DSA), metrics, Regulators, Safety Case Framework (SCF), safety management system, Verification and Validation (V&V)

A Reliability Analysis of Self-Driving Vehicles: Evaluating the Safety and Performance of Autonomous Driving Systems

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Item Type</b>         | Conference Paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Author</b>            | A. Pradeep                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Author</b>            | M. Bakoev                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Author</b>            | N. Akhroljonova                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Abstract</b>          | Self-driving cars are a ground-breaking invention with the potential to revolutionize the transportation sector. As technology develops, it is more crucial than ever to guarantee the dependability of self-driving cars. The dependability analysis of self-driving cars is the main topic of this research article. The multiple levels of automation in self-driving cars and their accompanying reliability requirements are covered in the first section of the study. Afterwards, it examines the many parts of a self-driving car system, such as the perception, decision-making, and control subsystems, and talks about the dependability issues that each of these parts faces. The paper ends by highlighting the value of reliability analysis in ensuring the security and wide acceptance of self-driving cars. In order to overcome the remaining issues and provide more sophisticated methods for self-driving car dependability evaluations, the report advocates for additional research. |
| <b>Date</b>              | 29-30 June 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>DOI</b>               | 10.1109/ECAI58194.2023.10194188                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Extra</b>             | Journal Abbreviation: 2023 15th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Pages</b>             | 1-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Proceedings Title</b> | 2023 15th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Conference Name</b>   | 2023 15th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Date Added</b>        | 20/09/2026, 7:46:33 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Modified</b>          | 20/09/2026, 7:46:33 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Tags:**

perception, Safety, safety, self-driving cars, Transportation, Regulation, automation levels, control, decision-making, reliability analysis, Roads, Autonomous automobiles, Industries, Government

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## A Review on Edge Computing: Security Considerations and Challenges

**Item Type** Conference Paper

**Author** R. P. Singh

**Author** H. Barhaiya

**Author** S. Sahu

**Author** J. Kumar

**Author** D. S. Rathore

**Abstract** The foundation of Edge Computing revolves around the redistribution of computational tasks to leverage the currently underutilized computational capacities present at the network's edge. Particularly in the privacy and security industry, the expansion of cloud, IoT, and edge computing models raises serious concerns regarding data privacy and its protection from unauthorized access. However, adequate data privacy solutions for the privacy and security industry require thorough research. This document focuses on the security challenges inherent in edge computing, analysing the solutions researchers have proposed to mitigate these security concerns. Additionally, it provides a comprehensive review of cloud and edge computing, including a detailed comparison of current research, characteristics, requirements for enabling edge computing, and the associated challenges. The discourse further includes considerations for establishing a secure edge computing architecture, presenting various recommended solutions.

**Date** 29-30 Nov. 2024

**DOI** 10.1109/ICDCC62744.2024.10961646

**Extra** Journal Abbreviation: 2024 First International Conference on Data, Computation and Communication (ICDCC)

**Pages** 605-609

**Proceedings Title** 2024 First International Conference on Data, Computation and Communication (ICDCC)

**Conference Name** 2024 First International Conference on Data, Computation and Communication (ICDCC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Security, Data privacy, Computer architecture, IoT, Edge computing, Edge Computing, Reviews, Cloud computing, Industries, Protection, Data models, Data centers, Data Centers, Edge devices, Network

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## A Study on Analyzing the Security and Privacy Implications of Existing and Emerging Sensors in Autonomous Electric Vehicles

**Item Type** Journal Article

**Author** A. Shankar

**Author** C. Maple

**Author** G. Epiphaniou

**Abstract** The popularity of anonymous consumers' modern Electric Vehicles (EVs) has developed tremendously. The increasing demand for more environmentally friendly and resource-conserving transit systems has paved the way for the widespread acceptance of EVs. The EV uses the information communication technology to transmit information which ensures the effective transportation. The paper's analysis includes security issues regarding the Internet of vehicles, EV charging, sensors, anomaly detection, and vulnerable cyber-attacks. The analysis uses a wide range of datasets with different observations and statistical results and reports the greatest risk of privacy issues occurring in EVs communication. The major issues of security and privacy of autonomous electric vehicles within ecological settings are discussed, as well as a wide range of security risks, approaches, countermeasures, and solutions to handle them.

**Date** August 2025

**DOI** 10.26599/TST.2023.9010132

**Volume** 30

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**Journal Abbr** Tsinghua Science and Technology

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## Tags:

Safety, Security, Data privacy, Automobiles, Privacy, Transportation, Electric vehicle charging, Supply chains, cyber-attacks, Sensors, anomaly detection, Autonomous Electric Vehicles (AEVs), Internet of Vehicles, Internet of Vehicles (IoVs), security threats

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## A study on cyclists' risk-taking behavior intention on shared roads with autonomous vehicles

**Item Type** Conference Paper

**Author** S. Fu

**Author** H. Zhou

**Author** Y. Li

**Author** W. Wang

**Abstract** Automatic vehicles are likely to operate in mixed traffic conditions, affecting cyclists' existing behavioral decisions. However, the intention and decision-making mechanism of cyclists' risk-taking behavior on shared roads are not clearly understood. The theory of planned behavior(TPB) has been proven to be effective in the study of travel behavior. Based on the theory of TPB, the Extended Theory of Planned Behavior (ETPB) research framework of risk-taking behavior intention was constructed by introducing safety perception, psychological motivation and technology cognition variables. Safety perception variables took into account the public's perception of safety in the shared road environment with AVs, while psychological motivation mainly took into account

riders' past riding habits, risk perception bias and riding emotion factors. Secondly, questionnaire survey was used to obtain riders' personal basic information and multi-situation risk-taking behavior willingness data, and corresponding structural equation model was constructed to explain the influence of variables on willingness. Finally, clustering people by means of K-means clustering algorithm and analyzing the psychological and behavioral characteristics of all kinds of people, so as to facilitate the selection of traffic management strategies. The results show that the extended TPB model can better explain the risk-taking behaviors of cyclists on the self-driving shared road. The influence of attitude and safety perception on the willingness to take risks is more prominent, and technology cognition plays an important indirect role. This study expands the research on the psychology of cyclists' risk-taking behavior, which could help to promote the integration of autonomous vehicles into transportation systems and better management strategies.

**Date** 4-6 Aug. 2023

**DOI** 10.1109/ICTIS60134.2023.10243988

**Extra** Journal Abbreviation: 2023 7th International Conference on Transportation Information and Safety (ICTIS)

**ISBN** 2832-899X

**Pages** 1236-1242

**Proceedings Title** 2023 7th International Conference on Transportation Information and Safety (ICTIS)

**Conference Name** 2023 7th International Conference on Transportation Information and Safety (ICTIS)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

traffic safety, Transportation, Surveys, Psychology, Roads, Mathematical models, Behavioral sciences, automatic vehicles, Cognition, extended planning behavior theory, risky riding behavior, shared road

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## A Summary Of The Development Of Autonomous Vehicle

**Item Type** Conference Paper

**Author** H. Wu

**Author** A. D. Calderon

**Abstract** The emergence of automatic driving can improve transportation efficiency, improve travel safety, and achieve fast, safe and efficient travel. However, the research progress of autonomous vehicle in various countries is different. In order to help researchers understand the research status, development trends, hot issues and existing research results in this field, so as to improve research efficiency, this paper introduces the development status of autonomous vehicle. Based on the research background and the development status of autonomous vehicle technology, this paper briefly describes the development status of automated driving technology testing in China, and selects four typical countries and regions (the United States, Europe, Japan and China) to introduce the development status of technology. In addition, this paper analyzes the challenges encountered in the development of autonomous vehicle from five aspects: safety, laws and

regulations, ethics, privacy protection and social acceptance, and summarizes the possible solutions. The research shows that governments and technology companies of all countries have a strong interest in autonomous vehicle and have invested a lot of resources in research. At present, this field is developing very fast, and the problems caused by it will be overcome one by one.

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|--------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Date</b>              | 3-5 Nov. 2023                                                                                                  |
| <b>DOI</b>               | 10.1109/RCAE59706.2023.10398804                                                                                |
| <b>Extra</b>             | Journal Abbreviation: 2023 6th International Conference on Robotics, Control and Automation Engineering (RCAE) |
| <b>Pages</b>             | 174-180                                                                                                        |
| <b>Proceedings Title</b> | 2023 6th International Conference on Robotics, Control and Automation Engineering (RCAE)                       |
| <b>Conference Name</b>   | 2023 6th International Conference on Robotics, Control and Automation Engineering (RCAE)                       |
| <b>Date Added</b>        | 20/09/2026, 7:46:33 pm                                                                                         |
| <b>Modified</b>          | 20/09/2026, 7:46:33 pm                                                                                         |

**Tags:**

Autonomous vehicles, Safety, Privacy, Transportation, Regulation, autonomous driving, Europe, challenge, current situation, Government

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## A Survey on the Role of Artificial Intelligence and Machine Learning in 6G-V2X Applications

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|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Item Type</b> | Conference Paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Author</b>    | D. Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Author</b>    | A. Qiu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Author</b>    | Q. Zhou                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Author</b>    | H. D. Schotten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Abstract</b>  | The rapid advancement of Vehicle-to-Everything (V2X) communication is transforming Intelligent Transportation Systems (ITS), with 6G networks expected to provide ultra-reliable, low-latency, and high-capacity connectivity for Connected and Autonomous Vehicles (CAVs). Artificial Intelligence (AI) and Machine Learning (ML) have emerged as key enablers in optimizing V2X communication by enhancing network management, predictive analytics, security, and cooperative driving due to their outstanding performance across various domains, such as natural language processing and computer vision. This survey comprehensively reviews recent advances in AI and ML models applied to 6G-V2X communication. It focuses on state-of-the-art techniques, including Deep Learning (DL), Reinforcement Learning (RL), Generative Learning (GL) and Federated Learning (FL), with particular emphasis on developments from the past two years. Notably, AI, especially GL, has shown remarkable progress and emerging potential in enhancing the performance, adaptability, and intelligence of 6G-V2X systems. Despite these advances, a systematic summary of recent research efforts in this area remains lacking, which this survey aims to address. We analyze their roles in 6G-V2X applications such as intelligent resource allocation, beamforming, intelligent traffic management, and security |

management. Furthermore, we explore the technical challenges, including computational complexity, data privacy, and real-time decision-making constraints, while identifying future research directions for AI-driven 6G-V2X development. This study aims to provide valuable insights for researchers, engineers, and policymakers working towards realizing intelligent, AI-powered V2X ecosystems in 6G communication.

**Date** 27-30 Oct. 2025

**DOI** 10.1109/WF-IoT64238.2025.11270555

**Extra** Journal Abbreviation: 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

**ISBN** 2768-1734

**Pages** 1-7

**Proceedings Title** 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

**Conference Name** 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Artificial intelligence, V2X, AI, Decision making, Surveys, Vehicle-to-everything, Reviews, LLM, Biological system modeling, Real-time systems, Resource management, 6G, 6G mobile communication, FL, ML

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## A Zero-Trust Empowered Continuous Authentication System for Drivers: Integrating Federated Learning with Conditional Transformer GAN

**Item Type** Conference Paper

**Author** L. Ge

**Author** Z. Cao

**Author** M. Jiang

**Author** L. Zhao

**Abstract** To address the challenges of data insufficiency, model complexity, privacy protection, and zero-trust continuous authentication in intelligent connected vehicle driver identification, this paper proposes a Zero-Trust Driver Continuous Authentication System that integrates Federated Learning with Conditional Transformer GAN (FCG). FCG employs a structurally concise Memory-Augmented Autoencoder to achieve efficient feature learning, and proposes a Transformer-based GAN model for data augmentation. It innovatively utilizes existing sensor data as conditional input to generate cross-physical quantity data, which both alleviates the data scarcity problem and preserves the natural correlations between different data types. For privacy protection, FCG combines the Memory-Augmented Autoencoder with a federated learning framework, achieving distributed collaborative training where 'models move instead of data,' while simultaneously meeting the requirements of continuous verification under the zero-trust security model. Extensive experiments on authoritative datasets demonstrate that FCG significantly outperforms existing methods in terms of false alarm rate and authentication accuracy. The research results of this paper are expected to promote the development and application of identity authentication

technology for intelligent connected vehicles.

**Date** 14-17 Nov. 2025

**DOI** 10.1109/Trustcom66490.2025.00115

**Extra** Journal Abbreviation: 2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)

**ISBN** 2324-9013

**Pages** 996-1006

**Proceedings Title** 2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)

**Conference Name** 2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)

**Date Added** 20/09/2026, 7:46:34 pm

**Modified** 20/09/2026, 7:46:34 pm

### Tags:

Security, Data privacy, Authentication, Privacy, Federated learning, Connected vehicles, Federated Learning, Protection, Transformers, Data models, Training, Anomaly Detection, Conditional Transformer GAN, Privacy Protection Framework

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## Active Directory-Inspired Identity and Access Management Framework for Secure Connected Vehicle Ecosystems

**Item Type** Conference Paper

**Author** M. Seyfipoor

**Author** M. M. Eskandari

**Author** S. Mohammadi

**Abstract** Connected and autonomous vehicles (CAVs) rely on continuous cloud–edge–vehicle communication, making identity and access management a critical security challenge. Traditional solutions such as PKI or blockchain-based authentication provide integrity but lack a unified, hierarchical trust model. This paper introduces the Vehicle Directory Service (VDS), a lightweight, Active Directory-inspired framework for secure credential issuance and authentication in connected vehicle ecosystems. VDS extends the Kerberos ticket-based mechanism to a distributed Cloud–Edge–Vehicle hierarchy, enabling short-lived, context-bound credentials with low computation and latency overhead. Experimental evaluation on a SUMO–NS3 testbed demonstrates that VDS achieves 96.7% authentication success rate, reduces authentication latency by 68%, and maintains less than 8% CPU utilization, outperforming existing centralized and blockchain-based models. The proposed framework establishes a scalable, low-latency, and secure identity foundation for next-generation intelligent transportation networks.

**Date** 24-25 Dec. 2025

**DOI** 10.1109/ICSPIS68676.2025.11551705

**Extra** Journal Abbreviation: 2025 11th International Conference on Signal Processing and Intelligent Systems (ICSPIS)

**Pages** 102-107

**Proceedings Title** 2025 11th International Conference on Signal Processing and Intelligent Systems (ICSPIS)  
**Conference Name** 2025 11th International Conference on Signal Processing and Intelligent Systems (ICSPIS)  
**Date Added** 20/09/2026, 7:46:33 pm  
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**Tags:**

Security, Vehicles, Authentication, Signal detection, Connected vehicles, Connected Vehicles, Modeling, Active Directory, Cloud–Edge Architecture, Clouds, Joining processes, Kerberos Authentication, Lightweight Security, Management, Printing, Vehicle Directory Service (VDS)

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## Adopting linear optimization to support autonomous vehicles in smart city

**Item Type** Conference Paper  
**Author** N. Petrovic  
**Author** D. Kocic  
**Abstract** In this paper, we consider leveraging linear optimization technique to cover two aspects within smart city architecture related to autonomous vehicles – vehicle battery charging and sensor data processing. Moreover, a tool for modeling and simulation of smart city scenarios involving autonomous vehicles and elements of edge computing is presented and integrated with the proposed linear optimization models for evaluation purposes.  
**Date** 26-27 Nov. 2019  
**DOI** 10.1109/TELFOR48224.2019.8971185  
**Extra** Journal Abbreviation: 2019 27th Telecommunications Forum (TELFOR)  
**Pages** 1-4  
**Proceedings Title** 2019 27th Telecommunications Forum (TELFOR)  
**Conference Name** 2019 27th Telecommunications Forum (TELFOR)  
**Date Added** 20/09/2026, 7:46:33 pm  
**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

Autonomous vehicles, simulation, modeling, Computer architecture, smart city, Edge computing, Smart cities, edge computing, Telecommunications, Computational modeling, Batteries, Data processing, linear optimization, Optimization, Optimization models

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## Advanced Autonomous Vehicle Acceptance Model (AAVAM) for Promoting Accessible Travel

**Item Type** Conference Paper  
**Author** L. Chen

**Author** L. Yang

**Author** X. Luan

**Author** L. Yang

**Author** Y. Song

**Author** S. Qin

**Abstract** The rise of autonomous driving technology offers new opportunities for accessible travel for the elderly population. However, to ensure that this technology meets the specific needs of the elderly, it is essential to understand their acceptance of autonomous vehicles and the factors that influence it. This study proposes the Advanced Autonomous Vehicle Acceptance Model (AAVAM) to systematically explore the acceptance of autonomous vehicles by the elderly at different stages of the vehicle's life cycle (before purchase, during use, and at the end of life). By comprehensively considering factors from product, service, and system aspects, the AAVAM model provides a theoretical basis for designing and providing autonomous vehicle solutions that meet the needs of the elderly.

**Date** 28-30 Aug. 2024

**DOI** 10.1109/ICAC61394.2024.10718798

**Extra** Journal Abbreviation: 2024 29th International Conference on Automation and Computing (ICAC)

**Pages** 1-6

**Proceedings Title** 2024 29th International Conference on Automation and Computing (ICAC)

**Conference Name** 2024 29th International Conference on Automation and Computing (ICAC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous vehicles, Automation, Older adults, Computational modeling, acceptance model, elderly population, social inclusion

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## An Intelligent Terminal Based Privacy-Preserving Multi-Modal Implicit Authentication Protocol for Internet of Connected Vehicles

**Item Type** Journal Article

**Author** F. Wei

**Author** S. Zeadally

**Author** P. Vijayakumar

**Author** N. Kumar

**Author** D. He

**Abstract** The Internet of connected Vehicles (IOV) can collect, process, compute and release the information of intelligent transportation systems. IOV is an integrated service system that can support the applications for automatic driving, intelligent transport and information services. As the number of incidents on IOV has been on the rise in the past few years, IOV security is becoming increasingly important in the IOV architecture. One of the most notable risks of IOV faces is intelligent terminal security. The vehicle's intelligent terminal can be used to launch for further attacks on the on-board operating system to penetrate into the internal network of connected

vehicle, and consequently threaten the safety of the vehicle. Thus, it is of paramount importance that we protect the security of the intelligent terminal. We propose two intelligent terminal based privacy-preserving multi-modal implicit authentication protocols to protect the security of the intelligent terminal in IOV. The proposed protocols use the password and the vehicle owner's behavior features as the authentication factors to protect the security of the intelligent terminal. Since the vehicle owner's behavior features are sensitive and the privacy information of the user must be protected, we also consider the privacy protection of the behavior features. Our protocols do not reveal any information about the vehicle owner's behavior features to the authentication server and the adversary except the ciphertext size of the feature vector. We analyze the security of our proposed protocol and compare them with other related protocols in terms of computation and communications costs. Our results demonstrate that our proposed protocols yield better security and efficiency.

**Date** July 2021

**DOI** 10.1109/TITS.2020.2998775

**Volume** 22

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**Date Added** 20/09/2026, 7:46:33 pm

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### Tags:

Authentication, Privacy, privacy, connected vehicles, Connected vehicles, protocol, Protocols, Internet of Vehicles, intelligent terminal, Internet, Password

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## An Overview of Privacy-Preserving and Security Methods for Deep Learning

**Item Type** Conference Paper

**Author** P. K. Goel

**Author** S. Verma

**Author** L. S. Mahur

**Author** A. Aeron

**Author** C. Jayant

**Author** K. Komal

**Abstract** The advent of deep learning has revolutionized diverse domains, showcasing remarkable performance in tasks such as image recognition, natural language processing, and autonomous systems. Despite its widespread adoption and myriad advantages, concerns about privacy and security loom large, especially when handling sensitive data. This research paper surveys the challenges, proposes countermeasures, and sheds light on open issues and potential threats within the industry. We present a comprehensive overview of privacy-preserving and security methods for deep learning, addressing the deployment challenges in privacy-sensitive applications.

**Date** 7-8 June 2024  
**DOI** 10.1109/ICICET59348.2024.10616340

**Extra** Journal Abbreviation: 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

**Pages** 1-6

**Proceedings Title** 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

**Conference Name** 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

Data privacy, Deep learning, Privacy, Technological innovation, privacy concerns, privacy-preserving methods, natural language processing, Protocols, Computational modeling, autonomous systems, Training, image recognition, potential threats, security challenges, security methods, sensitive data

Analyzing the inconsistency in driving patterns between manual and autonomous modes under complex driving scenarios with a VR-enabled simulation platform

**Item Type** Journal Article

**Author** Z. Xu

**Author** Y. Fang

**Author** N. Zheng

**Author** H. L. Vu

**Abstract** Purpose - With the aid of naturalistic simulations, this paper aims to investigate human behavior during manual and autonomous driving modes in complex scenarios. Design/methodology/approach - The simulation environment is established by integrating virtual reality interface with a micro-simulation model. In the simulation, the vehicle autonomy is developed by a framework that integrates artificial neural networks and genetic algorithms. Human-subject experiments are carried, and participants are asked to virtually sit in the developed autonomous vehicle (AV) that allows for both human driving and autopilot functions within a mixed traffic environment. Findings - Not surprisingly, the inconsistency is identified between two driving modes, in which the AV's driving maneuver causes the cognitive bias and makes participants feel unsafe. Even though only a shallow portion of the cases that the AV ended up with an accident during the testing stage, participants still frequently intervened during the AV operation. On a similar note, even though the statistical results reflect that the AV drives under perceived high-risk conditions, rarely an actual crash can happen. This suggests that the classic safety surrogate measurement, e.g. time-to-collision, may require adjustment for the mixed traffic flow. Research limitations/implications - Understanding the behavior of AVs and the behavioral difference between AVs and human drivers are important, where the developed platform is only the first effort to identify the critical scenarios where the AVs might fail to react. Practical implications - This paper attempts to fill the existing research gap in preparing close-to-reality tools for AV experience and further understanding

human behavior during high-level autonomous driving. Social implications - This work aims to systematically analyze the inconsistency in driving patterns between manual and autopilot modes in various driving scenarios (i.e. multiple scenes and various traffic conditions) to facilitate user acceptance of AV technology. Originality/value - A close-to-reality tool for AV experience and AV-related behavioral study. A systematic analysis in relation to the inconsistency in driving patterns between manual and autonomous driving. A foundation for identifying the critical scenarios where the AVs might fail to react.

**Date** 2022

**DOI** 10.1108/JICV-05-2022-0017

**Volume** 5

**Pages** 215-234

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**ISSN** 2399-9802

**Issue** 3

**Journal Abbr** Journal of Intelligent and Connected Vehicles

**Date Added** 20/09/2026, 7:46:32 pm

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### Tags:

Autonomous vehicles, Accidents, Virtual reality, Autonomous vehicle, Sensors, Vehicle dynamics, Behavioral sciences, Cognitive bias, Intervention behavior, Manuals, Mixed traffic, Naturalistic simulation, Solid modeling

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## Artificial Intelligence Governance: A Study on the Ethical and Security Issues that Arise

**Item Type** Conference Paper

**Author** A. Ndrejaj

**Author** M. Ali

**Abstract** artificial intelligence (AI) is rapidly becoming integrated into every facet of society. However, its use in current governance is lacking. Across major global powers, attempts are currently being made to regulate AI with the aim of protecting citizens, corporations and political realms. Despite this, there have been several serious incidents involving AI recently. The current gaps in governance could give way to even greater future problems as AI becomes a more pervasive part of society. This research examines both current attempts at governance and common concerns to address using both a case study on Amazon's Alexa and a previously conducted survey of U.S. public opinion on AI. The results of these studies show that AI is not necessarily always secure and that citizens' concerns over AI risks are growing. Overall, this research highlights some of the key areas governing bodies should seek to address.

**Date** 17-18 Aug. 2022

**DOI** 10.1109/iCCECE55162.2022.9875082

**Extra** Journal Abbreviation: 2022 International Conference on Computing, Electronics & Communications Engineering (iCCECE)

**Pages** 104-111  
**Proceedings Title** 2022 International Conference on Computing, Electronics & Communications Engineering (iCCECE)  
**Conference Name** 2022 International Conference on Computing, Electronics & Communications Engineering (iCCECE)  
**Date Added** 20/09/2026, 7:46:33 pm  
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### Tags:

Security, Artificial intelligence, AI, artificial intelligence, Privacy, security, governance, survey, Law, Ethics, case study, Minimization, public opinion, Virtual assistants

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## Assessing Public Acceptance of Autonomous Vehicles in Saudi Arabia

**Item Type** Conference Paper  
**Author** A. Alahmadi  
**Author** S. Elhag  
**Author** M. Meccawy  
**Abstract** Advancements in Artificial Intelligence (AI) are changing various sectors, and the transportation sector is a key area for innovation. Autonomous Vehicles (AVs) utilize AI technologies such as deep learning, computer vision, and image processing. As these vehicles become increasingly important for future mobility, assessing their implementation readiness becomes essential. Public acceptance of AVs plays a critical role in shaping policies and strategies for the successful integration of AVs into the transportation ecosystem in the future. This research aims to assess public acceptance of AVs, which is critical for their successful integration in the early stages. A conceptual model has been designed based on theoretical models. The research employed both quantitative and qualitative research methods. The primary method of data collection was an online survey conducted in Saudi Arabia. The sample comprised 112 males and 96 females, demonstrating a fair representation of both genders in the study. The results show that perceived usefulness, performance expectancy, and cost were among the most significant factors of adoption intention. Additionally, the participants' primary concerns about AVs include malfunctions, infrastructure limitations, as well as security and privacy concerns. The results underscore the need for public participation, especially in the early stages, to shape AV acceptance in the region. This offers valuable guidance for policymakers and transport stakeholders working to facilitate future adoption.  
**Date** 19-22 Oct. 2025  
**DOI** 10.1109/AICCSA66935.2025.11315380  
**Extra** Journal Abbreviation: 2025 IEEE/ACS 22nd International Conference on Computer Systems and Applications (AICCSA)  
**ISBN** 2161-5330  
**Pages** 1-7  
**Proceedings Title** 2025 IEEE/ACS 22nd International Conference on Computer Systems and Applications (AICCSA)

**Conference Name** 2025 IEEE/ACS 22nd International Conference on Computer Systems and Applications (AICCSA)  
**Date Added** 20/09/2026, 7:46:32 pm  
**Modified** 20/09/2026, 7:46:32 pm

### Tags:

Autonomous vehicles, Security, Artificial intelligence, Transportation, Technological innovation, Costs, Public Acceptance, Surveys, Technology Acceptance, Autonomous Vehicles (AVs) Acceptance, Future Transportation, Smart Mobility, Robustness, Shape, Stakeholders

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## Assessing the Security and Architectural Frameworks of Self-Driving Cars in Smart Cities

**Item Type** Conference Paper

**Author** M. J. C. Samonte

**Author** L. M. D. Laqueo

**Author** R. K. L. Mampusti

**Author** K. D. H. Panganiban

**Abstract** This review aims at presenting a clear understanding of AVs in smart cities, more specifically in terms of technological development, inherent challenges, and social impacts. LiDAR sensors and the communication protocols essential for improving the navigation and safety of AVs in urban areas are explored. The review brings together recent studies and notes the resultant adversities of the AV sensor system with respect to the weather impact as well as the need to strengthen protection against cyber-crimes. Other key areas of concern include road traffic control, traffic flow reduction, energy conservation, and changes in the transport system usage within urban areas. In incorporating various aspects related to the AV integration, the review highlights the role of policy makers, researchers and industry players, and stresses the need for regulatory framework, societal acceptance, and design of interaction between human and computers as crucial in the development of AVs. AVs hold the potential of enabling smarter, safer, and sustainable cities worldwide in the future and the consequent growth of sensor technology and communication networks for AV technologies are prospective for urban mobility.

**Date** 25-27 Oct. 2024

**DOI** 10.1109/ICSGSC62639.2024.10813654

**Extra** Journal Abbreviation: 2024 8th International Conference on Smart Grid and Smart Cities (ICSGSC)

**ISBN** 2768-0088

**Pages** 374-381

**Proceedings Title** 2024 8th International Conference on Smart Grid and Smart Cities (ICSGSC)

**Conference Name** 2024 8th International Conference on Smart Grid and Smart Cities (ICSGSC)

**Date Added** 20/09/2026, 7:46:33 pm

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### Tags:

autonomous vehicles, Security, smart cities, Smart cities, AV integration, cybersecurity risks, System Architecture, Reviews, Stress, Protocols, Protection, Intelligent sensors, Road traffic control, Sensor systems, Smart grids

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## Autonomous mobility and user perception: A Case of City as a Lab in Slovenia

**Item Type** Conference Paper

**Author** I. Zajc

**Author** R. Sernec

**Author** G. Lenart

**Author** A. Pucihar

**Abstract** Autonomous vehicles (AV) have the potential to disrupt the entire transport industry. Technology of AV offers many opportunities as for example reduction of road accidents, fewer congestions on the roads, lower number of vehicles while having better utilization. Beside significant contribution of electric vehicles (EV) to cleaner environment, fully AV also bring new social element as they enable mobility for all. In addition, the use of digital technologies in combination with AV introduces a radical and new paradigm in the transportation, where the lines between car ownership, rental, and lease modes are more and more blurred. To explore the potential of AV in smart city context, AV Living Lab has been established on the premises of BTC City. The AV Living lab was created to test and to learn about real-life solutions for implementation of AV. The underlying concept is BTC City as a Lab ecosystem, where the latest advanced technologies, business models, and services are tested with real users, real cars, on real roads over the real interactions in cross-industry environment. We describe a specific use case of AV shuttle driving and its human experience perception, which was evaluated after a large scale pilot demonstration.

**Date** 28 Sept.-2 Oct. 2020

**DOI** 10.23919/MIPRO48935.2020.9245147

**Extra** Journal Abbreviation: 2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)

**ISBN** 2623-8764

**Pages** 1843-1847

**Proceedings Title** 2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)

**Conference Name** 2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Automobiles, Transportation, Legislation, Smart cities, autonomous mobility, Roads, case, city as a lab, living lab, Road accidents, Slovenia

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# Autonomous Vehicle and its Adoption: Challenges, Opportunities, and Future Implications

**Item Type** Conference Paper

**Author** M. Ahmed

**Author** R. Iqbal

**Author** S. Amin

**Author** O. Alhabshneh

**Author** A. Garba

**Abstract** The future of mobility and adoption by the public has become a matter of general concern for commuters, drivers, automotive industry and public authorities. Future transport is expected to be dominated by self-driving and autonomous vehicles. Our extensive studies on user adoption indicates that acceptance will be determined mainly by these concerns. In this paper, we investigated user adoption and acceptance of self-driving vehicles. Our study demonstrated that the future adoption of self-driving vehicles will be affected by several inherent concerns such as security, privacy, and trust by the public. Nevertheless, the results of our review revealed that the field of self-driving vehicles will continue to generate interest due to its impact on transport and mobility. Predicting user adoption of self-driving vehicles is a novel research subject which has recently begun to gather momentum; however, predictive accuracy remains a challenge using simple statistical methods. Machine learning modelling techniques can provide better understanding of data and non-linear relationships between decision variables. From our data, we were able to predict self-driving acceptance based on user preference and inherent concerns. We applied supervised learning algorithms namely, Naïve Bayes, Random Forest, and Fuzzy Logic to predict user adoption of self-driving vehicles. We used 6 independent variables namely safety, trust, security, ethics, cost, and privacy. We evaluated each algorithm by using 5-fold cross validation technique. The algorithms were compared based on 3 outcomes: accuracy, precision and recall. Fuzzy logic was found to outperform other algorithms followed by random forest while Naïve Bayes performed lower.

**Date** 23-24 Nov. 2022

**DOI** 10.1109/ETCEA57049.2022.10009804

**Extra** Journal Abbreviation: 2022 International Conference on Emerging Trends in Computing and Engineering Applications (ETCEA)

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**Proceedings Title** 2022 International Conference on Emerging Trends in Computing and Engineering Applications (ETCEA)

**Conference Name** 2022 International Conference on Emerging Trends in Computing and Engineering Applications (ETCEA)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

autonomous vehicles, Safety, Security, Privacy, Supervised learning, user acceptance, Prediction algorithms, Statistical analysis, fuzzy logic, Fuzzy logic, predictive modelling

## Autonomous Vehicles - Trust, Safety and Security Cases: The Complete Picture

**Item Type** Conference Paper

**Author** T. Myklebust

**Author** T. Stålhane

**Author** G. D. Jenssen

**Abstract** Summary & Conclusions Safety cases are required by several functional safety standards, specifications, and guidelines. Cybersecurity cases have recently been required by ISO/SAE 21424:2021 for automotive and EN TS 50701:2021 for the railway domain. In this paper we discuss cybersecurity cases and suggest using the topics and structure for a cybersecurity case as described in Annex G of EN TS 50701. BSI PAS 1881:2022 requires: "Trialing organizations shall develop and publish a publicly available and accessible version of the safety case". We have already developed a "safety case for the public" [1] to ensure that (1) the public is aware that safety evidence exists, (2) they are aware of relevant safety aspects when they are passengers, and (3) the vehicle's limitations are described transparently. Trust is a dynamic process that involves initiating and building trust, responding to violations of trust (failures), and trying to rebuild (repair) trust. The building blocks of trust are not limited to the vehicle itself but also include the embedded AI (Artificial Intelligence) and its overt function. Trust is a holistic perception of the complete service, technology, and organizations responsible for developing, implementing, and certifying an autonomous vehicle. An autonomous vehicle will need acceptance from the certification bodies and the authorities, but we also need to gain the public's trust. Our research found that several aspects are missing in the safety and cybersecurity cases to ensure public trust. To make self-driving buses a success, they need to be considered trustworthy. Thus, we need a "Trust case" that includes evidence related to distinct trust aspects. Our literature studies, focus groups [4], and surveys found that trust and safety are not correlated. We have developed a "Trust case" to cover the factors not included in the safety and cybersecurity cases. The resulting "Trust case" approach is currently in the form of specific information topics presented in a layman form and a safety case for the public [6], and specific trust topics in [7]. Further research is necessary, related to topics such as deep learning, security, and incorrect reporting to the driver due to e.g., false positive results.

**Date** 23-26 Jan. 2023

**DOI** 10.1109/RAMS51473.2023.10088202

**Extra** Journal Abbreviation: 2023 Annual Reliability and Maintainability Symposium (RAMS)

**ISBN** 2577-0993

**Pages** 1-6

**Proceedings Title** 2023 Annual Reliability and Maintainability Symposium (RAMS)

**Conference Name** 2023 Annual Reliability and Maintainability Symposium (RAMS)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Safety, Artificial intelligence, safety, Reliability, Safety case, autonomous, public, Computer security, cybersecurity case, Organizations, Random access memory, Standards organizations, trust case

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## Autonomous Vehicles: Building a Test-bed Prototype at a Controlled Environment

**Item Type** Conference Paper

**Author** S. El-Tawab

**Author** N. Sprague

**Author** A. Mufti

**Abstract** Rapid improvements in technology are making autonomous vehicles feasible. Researchers and industry are looking for different methods to implement the concept of smart cities. Autonomous Vehicles provide a unifying framework for several research topics (e.g., navigation, localization and Global Positioning System (GPS) positioning, security, and privacy). Despite significant progress, we believe that fully autonomous vehicles will be deployed initially in constrained environments (e.g., closed-loop areas, gated retirement communities). In this paper, we describe the development of a prototype autonomous golf-cart that can be used as a test-bed for a range of research projects. We explain the details of the conversion from a traditional golf-cart to an autonomously driven one. Our target application is a system of autonomously driven golf-carts for use within a retirement community. We show our testing results, and how Internet of Things (IoT) devices can be used to extend the WiFi coverage for log reporting and to request the golf-cart. Finally, we discuss the challenges and highlight improvements that can be performed on the system.

**Date** 2-16 June 2020

**DOI** 10.1109/WF-IoT48130.2020.9221222

**Extra** Journal Abbreviation: 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)

**Pages** 1-6

**Proceedings Title** 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)

**Conference Name** 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Security, Privacy, Smart cities, Internet of Things, Navigation, Internet of Things (IoT), Autonomous Vehicles (AV), Data Aggregation, Information Dissemination, Localization, Prototypes, Retirement, Test-bed Prototype

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## Autonomous vehicles: The future of automobiles

**Item Type** Conference Paper

**Author** M. V. Rajasekhar

**Author** A. K. Jaswal

**Abstract** Autonomous cars are the future smart cars anticipated to be driver less, efficient and crash avoiding ideal urban car of the future. To reach this goal automakers have started working in this area to realized the potential and solve the challenges

currently in this area to reach the expected outcome. In this regard the first challenge would be to customize and imbibe existing technology in conventional vehicle to translate them to a near expected autonomous car. This transition of conventional vehicles into an autonomous vehicle by adopting and implementing different upcoming technologies is discussed in this paper. This includes the objectives of autonomous vehicles and their implementation difficulties. The paper also touches upon the existing standards for the same and compares the introduction of autonomous vehicles in Indian market in comparison to other markets. There after the acceptance approach in Indian market scenarios is discussed for autonomous vehicles.

**Date** 27-29 Aug. 2015

**DOI** 10.1109/ITEC-India.2015.7386874

**Extra** Journal Abbreviation: 2015 IEEE International Transportation Electrification Conference (ITEC)

**Pages** 1-6

**Proceedings Title** 2015 IEEE International Transportation Electrification Conference (ITEC)

**Conference Name** 2015 IEEE International Transportation Electrification Conference (ITEC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Automobiles, Autonomous vehicle, Sensors, Mobile robots, Roads, Testing, Laser, Laser radar, Planning, RADAR

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## AVTAR Cyber Shield: A Comprehensive Framework for Machine Learning-Enhanced Cybersecurity in Cyber-Physical Environment

**Item Type** Conference Paper

**Author** V. N. K. Sykam

**Author** A. Goyal

**Author** M. Nijim

**Author** A. Mishra

**Author** D. Hicks

**Abstract** abstract-In the rapidly advancing era of self-driving cars, the imperative of ensuring robust cybersecurity measures to safeguard against evolving threats becomes paramount. This paper investigates the intricate cyber-physical systems within autonomous vehicles, emphasizing the critical role of “AVTAR Cyber Shield” as a comprehensive cybersecurity framework for fortifying against potential vulnerabilities. The paper provides a basic understanding on how autonomous car manufacturers and/or technology providers like Pony.ai, Waymo, WeRide and Zoox implement safety measures which prompted us to introduce a new driver’s self-preparation safety module before riding to qualitatively evaluate the preride risk(s) associated with self-driving cars. An exploration of current cybersecurity threats, and compliance with industry standards, is complemented by the identification of challenges such as technical complexities and human factors. The paper concludes with analyzing the existing statistical result focus on recent risk calculation, contributing to the future trends and innovations,

proposing recommendations that advocate for continuous monitoring, best practices, and adaptive strategies to address evolving cybersecurity threats and traditional human un/known mistakes. This comprehensive examination contributes valuable insights to the discourse on securing the future of self-driving vehicles, offering guidance for researchers, practitioners, and policymakers navigating the complex landscape of autonomous transportation.

**Date** 28-29 June 2024

**DOI** 10.1109/ICCSC62074.2024.10616984

**Extra** Journal Abbreviation: 2024 International Conference on Circuit, Systems and Communication (ICCSC)

**Pages** 1-6

**Proceedings Title** 2024 International Conference on Circuit, Systems and Communication (ICCSC)

**Conference Name** 2024 International Conference on Circuit, Systems and Communication (ICCSC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Safety, cybersecurity, self-driving cars, Transportation, attacks, countermeasures, cyber-physical systems, framework, risk mitigation, safety measures, threats, vulnerabilities, Autonomous automobiles, Industries, Real-time systems, Human factors, Time measurement

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## Blockchain-Based Internet of Vehicles (IoV): An Efficient Secure Ad Hoc Vehicular Networking Architecture

**Item Type** Conference Paper

**Author** S. Sharma

**Author** K. K. Ghanshala

**Author** S. Mohan

**Abstract** With the transformation of connected vehicles into the Internet of Vehicles (IoV), the time is now ripe for paving the way for the next generation of connected vehicles with novel applications and innovative security measures. The connected vehicles are experiencing prenominal growth in the auto industry, but are still studded with many security and privacy vulnerabilities. Today's IoV applications are part of cyber physical communication systems that collect useful information from thousands of smart sensors associated with the connected vehicles. The technology advancement has paved the way for connected vehicles to share significant information among drivers, auto manufacturers, auto insurance companies and operational and maintenance service providers for various applications. The critical issues in engineering the IoV applications are effective to use of the available spectrum and effective allocation of good channels an opportunistic manner to establish connectivity among vehicles, and the effective utilization of the infrastructure under various traffic conditions. Security and privacy in information sharing are the main concerns in a connected vehicle communication network. Blockchain technology facilitates secured communication among users in a connected vehicles network. Originally, blockchain technology was developed and employed with the cryptocurrency.

Bitcoin, to provide increased trust, reliability, and security among users based on peer-to-peer networks for transaction sharing. In this paper, we propose to integrate blockchain technology into ad hoc vehicular networking so that the vehicles can share network resources with increased trust, reliability, and security using distributed access control system and can benefit a wider scope of scalable IoV applications scenarios for decision making. The proposed architecture is the faithful environment for information sharing among connected vehicles. Blockchain technology allows multiple copies of data storage at the distribution cloud. Distributed access control system is significantly more secure than a traditional centralized system. This paper also describes how important of ad hoc vehicular networking in human life, possibilities in real-world implementation and its future trends. The ad hoc vehicular networking may become one of the most trendy networking concepts in the future that has the perspective to bring out much ease human beneficial and secured applications.

**Date** 30 Sept.-2 Oct. 2019

**DOI** 10.1109/5GWF.2019.8911664

**Extra** Journal Abbreviation: 2019 IEEE 2nd 5G World Forum (5GWF)

**Pages** 452-457

**Proceedings Title** 2019 IEEE 2nd 5G World Forum (5GWF)

**Conference Name** 2019 IEEE 2nd 5G World Forum (5GWF)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Security, Reliability, Blockchain, Privacy, privacy, security, smart city, Connected vehicles, blockchain, IoV, Intelligent sensors, intelligent sensors

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## Bridging Innovation and Inclusivity: User Perceptions of Autonomous Shuttles in Luxembourg

**Item Type** Conference Paper

**Author** M. Picone

**Author** F. Viti

**Author** A. Carteni

**Abstract** Autonomous shuttles are regarded as a concrete application of Intelligent Transport Systems (ITS) by combining automated driving with V2I communication and real-time monitoring, offering one of the most promising service innovations in public transport. Yet their large-scale deployment still faces significant technological and social challenges, and the impact of this technology to the perceived quality of service is still unclear. This study investigates user perceptions of an autonomous shuttle service operating in Luxembourg, based on a dedicated mobility survey. Two main findings emerge from the analysis. First, results highlight an ongoing transition phase toward the acceptance of this innovative autonomous driving technology. Although curiosity and interest in experiencing the service are evident, concerns about safety and reliability persist. In this context, the presence of an on-board supervisor emerges as a key factor in fostering user trust and improving service acceptability. Second, the survey data collection campaign revealed that the service is predominantly used by elderly

individuals and people with reduced mobility, suggesting that autonomous shuttles hold significant potential to improve accessibility for these user groups, and hence can contribute to promote inclusive urban mobility. These insights can guide the future development of autonomous public transport services, supporting the design of solutions more closely aligned with the needs of vulnerable user groups and more widely accepted by the public.

**Date** 8-10 Sept. 2025

**DOI** 10.1109/MT-ITS68460.2025.11223564

**Extra** Journal Abbreviation: 2025 9th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)

**Pages** 1-7

**Proceedings Title** 2025 9th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)

**Conference Name** 2025 9th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)

**Date Added** 20/09/2026, 7:46:34 pm

**Modified** 20/09/2026, 7:46:34 pm

### Tags:

Autonomous vehicles, Safety, Reliability, Technological innovation, Data collection, Urban areas, Uncertainty, Accessibility and social equity, Shared Autonomous Vehicles (SAVs), Sustainable and inclusive mobility, Users perceptions, Surveys, Older adults, Socioeconomics

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## Comparative Analysis of Regulatory Focus in Autonomous Vehicles (AVs) and Maritime Autonomous Surface Ships (MASS)

**Item Type** Conference Paper

**Author** T. E. Kim

**Author** J. -U. Schröder-Hinrichs

**Abstract** The increasing deployment of autonomous systems in transportation is reshaping how mobility systems operate and how regulatory frameworks must evolve to facilitate the shift from human-driven to AI-operated systems. This study presents a comparative analysis of regulatory discourse in two transport sectors at the frontier of automation: Autonomous Vehicles (AVs) and Maritime Autonomous Surface Ships (MASS). We used Natural Language Processing (NLP) and lifecycle phase mapping to extract and visualize the distribution of regulatory attention across key governance stages from standards development to testing, certification, liability and international harmonization. Gradient spectrum plots and semantic concept maps revealed significant asymmetries. These findings suggest a temporal lag in maritime governance and inform the importance of differentiated and phase-aware governance strategies in cross-sectoral technological transitions.

**Date** 7-10 Dec. 2025

**DOI** 10.1109/IEEM63636.2025.11357611

**Extra** Journal Abbreviation: 2025 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)

**Pages** 0738-0742**Proceedings Title** 2025 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)**Conference Name** 2025 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)**Date Added** 20/09/2026, 7:46:33 pm**Modified** 20/09/2026, 7:46:33 pm**Tags:**

Autonomous vehicles, Transportation, Technological innovation, Visualization, Natural language processing, Semantics, Autonomous Vehicles (AVs), Testing, Standards, autonomous systems, Marine vehicles, Maritime Autonomous Surface Ships (MASS), Surface treatment

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## Critical Factors Influencing Acceptance of Automated Vehicles by Hong Kong Drivers

**Item Type** Journal Article**Author** S. S. Man**Author** W. Xiong**Author** F. Chang**Author** A. H. S. Chan

**Abstract** This study aimed to identify critical factors that influence acceptance of automated vehicles among drivers. The focus of this study was on automated vehicles (AVs) of level 3 (conditional driving automation) as defined by the Society of Automotive Engineers. A research model was proposed here by using the technology acceptance model (TAM) with trust, risk perception (perceived safety risk and perceived privacy risk), compatibility, and system quality. A cross-sectional structured questionnaire survey was used to collect quantitative data from 237 drivers in Hong Kong. The data were analyzed to test the proposed research model by structural equation modeling. The proposed research model was found to explain 68% of the variance in intention to use AVs. In contrast with the TAM constructs of perceived usefulness and perceived ease of use, the results of this study indicated that trust was the most important factor in shaping a positive attitude towards using AVs, which affected driver intention to use AVs. Also, trust was found to be influenced by perceived safety risk, compatibility, and system quality. This study is the first attempt to consider technological factors related to AVs (compatibility and system quality) in explaining AV acceptance among drivers and highlighted the importance of the technological factors in the context of driver acceptance of AVs. Based on the findings of this study, several recommendations are discussed to help AV developers and governments to improve driver attitudes towards adoption of AVs.

**Date** 2020**DOI** 10.1109/ACCESS.2020.3001929**Volume** 8**Pages** 109845-109856**Publication** IEEE Access**ISSN** 2169-3536

**Journal Abbr** IEEE Access

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Safety, Vehicles, Automation, risk perception, trust, Acceptance, Privacy, Task analysis, Technology acceptance model, automated driving, compatibility, Modeling, system quality

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## Decentralized Defenses: A Framework for 5G IoT Security in Industry 4.0 using Blockchain Crowdsourcing

**Item Type** Conference Paper

**Author** P. Maiti

**Author** S. Prabhu

**Author** P. Ashok

**Abstract** This research work presents Blockchain Crowdsourcing-empowered 5G IoT as a backbone for network security of industry 4.0 in various applications. The study examines this aspect by featuring blockchain based crowdsourcing to detect and mitigate attacks. A structured literature search on the topic was conducted on various relevant bibliographic databases. 39 publications fulfilled the inclusion criteria. The study indicates that IoT devices in 5G ecosystems have serious security risk due to evolving threats and there are no proactive solutions. However, the best way to respond effectively is to work at the edge of the communication network using the density of IoT devices to obtain and analyze information, then taking corrective measures from the source of the threat. To the author's knowledge, this is the first study to apply the Blockchain based Crowdsourcing tool to mitigate cyber-attacks on 5G IoT networks for Industry 4.0. It has no single point of failure and caters to the increasing demand of cyber security workforce.

**Date** 13-14 March 2024

**DOI** 10.1109/ICSADL61749.2024.00093

**Extra** Journal Abbreviation: 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)

**Pages** 534-538

**Proceedings Title** 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)

**Conference Name** 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Cybersecurity, Network security, Blockchain, Sentiment analysis, IoT, Crowdsourcing, 5G, Information services, 5G mobile communication, 5G security, Blockchains, Digital witness, Healthcare 4.0, Industrial automation, Knowledge engineering, Proactive security, Smart contracts

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## Decentralized Identities for Enhanced Security in Vehicle-to-Everything (V2X)

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Item Type</b>         | Conference Paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Author</b>            | A. Kovačević                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Author</b>            | N. Gligorić                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Author</b>            | S. Jokić                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Abstract</b>          | The rise of Vehicle-to-Everything (V2X) communication in Intelligent Transportation Systems (ITS) enhances road safety and traffic management but introduces security vulnerabilities like impersonation and data tampering. Traditional identity management systems such as Public Key Infrastructure (PKI) are limited by their reliance on centralized authorities. This paper proposes a decentralized framework using Decentralized Identifiers (DIDs), Verifiable Credentials (VCs), and Distributed Ledger Technology (DLT), integrating Zero-Knowledge Proofs (ZKPs) to secure identity management in V2X systems. The proposed framework achieves robust authentication, mitigates centralization risks, and enables privacy-preserving mechanisms, demonstrating enhanced scalability and resilience against attacks in V2X communication. |
| <b>Date</b>              | 26-27 Nov. 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>DOI</b>               | 10.1109/TELFOR63250.2024.10819049                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Extra</b>             | Journal Abbreviation: 2024 32nd Telecommunications Forum (TELFOR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>ISBN</b>              | 2994-5828                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Pages</b>             | 1-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Proceedings Title</b> | 2024 32nd Telecommunications Forum (TELFOR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Conference Name</b>   | 2024 32nd Telecommunications Forum (TELFOR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Date Added</b>        | 20/09/2026, 7:46:32 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Modified</b>          | 20/09/2026, 7:46:32 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Tags:

Autonomous Vehicles, Road safety, Authentication, Privacy, Distributed ledger, Vehicle-to-Everything, Vehicle-to-everything, Decentralized Identifiers, Protocols, Public key, Resilience, Scalability, Telecommunications, Verifiable Credentials, Zero-Knowledge Proofs

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## Deep Learning-based Road Object Detection for Collision Avoidance in Autonomous Driving

|                  |                  |
|------------------|------------------|
| <b>Item Type</b> | Conference Paper |
| <b>Author</b>    | T. Sharma        |
| <b>Author</b>    | A. Chehri        |
| <b>Author</b>    | I. Fofana        |
| <b>Author</b>    | B. Debaque       |
| <b>Author</b>    | N. Duclos        |
| <b>Author</b>    | S. Khare         |

**Abstract** With the population in metro cities rising, traffic congestion has become a significant issue, resulting in an alarming number of traffic collisions and road accidents. Given the advancements in automated vehicles, it is crucial to have a vehicle detection system that is equipped to handle the various challenges that arise on the road, particularly when it comes to traffic collisions. Although many datasets are available to support object detection for traffic monitoring and management, the suitability of these datasets for specific weather conditions across the globe needs to be analyzed. We present Canadian vehicle datasets (CVD) and analyze the YOLOv8-based deep learning model using CVD. Thales, Canada, captured street-level videos in Quebec City, Canada, with a vehicle equipped with a high-quality RGB camera. The videos are taken day and night in all four seasons: rain, snow, hazy, and bright sunlight. We created 27378 labels for 11 classes for 8000 images. The YOLOv8 model trained on the publicly available roboFlow dataset was compared to the model trained on the combined roboflow and our weather-specific CVD.

**Date** 14-15 May 2024

**DOI** 10.1109/WFPST58552.2024.00020

**Extra** Journal Abbreviation: 2024 IEEE World Forum on Public Safety Technology (WFPST)

**Pages** 126-131

**Proceedings Title** 2024 IEEE World Forum on Public Safety Technology (WFPST)

**Conference Name** 2024 IEEE World Forum on Public Safety Technology (WFPST)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous Vehicles, Snow, Transportation, Object detection, Urban areas, Convolutional Neural Networks, Surveillance, Roads, Road accidents, Intelligent Transportation Systems, Sociology, YOLOv8

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## Designing an IoT Framework for Automated Driving Impact Analysis

**Item Type** Conference Paper

**Author** F. Bellotti

**Author** R. Berta

**Author** A. Kobeissi

**Author** N. Osman

**Author** E. Arnold

**Author** M. Dianati

**Author** B. Nagy

**Author** A. De Gloria

**Abstract** Automated Driving (AD) technology is rapidly advancing and being tested in large-scale pilots. The L3Pilot project will test AD functions at SAE automation level 3 and 4 in 100 cars, from 12 vehicle owners (car manufacturers and suppliers), with 1,000 drivers across 10 different countries in Europe. The project will thus generate a huge amount of data, part of which will be aggregated and

shared among project participants and publicly for answering a set of research questions about the evaluation of such aspects as technical and traffic issues, user acceptance, impact, socio-economic impact. This paper investigates the design of an Internet of Things (IoT) framework aimed at storing data collected from the prototype vehicles and pre-processed in the various test sites, in order to protect privacy and intellectual property (IP). The framework is expected to make such aggregated information easily available to analysts through a web interface. The framework, of which we present also a pre-pilot lab test, is accessible through a RESTful API and features different user roles to meet the different information access requirements.

**Date** 9-12 June 2019

**DOI** 10.1109/IVS.2019.8813989

**Extra** Journal Abbreviation: 2019 IEEE Intelligent Vehicles Symposium (IV)

**ISBN** 2642-7214

**Pages** 1111-1117

**Proceedings Title** 2019 IEEE Intelligent Vehicles Symposium (IV)

**Conference Name** 2019 IEEE Intelligent Vehicles Symposium (IV)

**Date Added** 20/09/2026, 7:46:34 pm

**Modified** 20/09/2026, 7:46:34 pm

### Tags:

Data privacy, Automation, Automobiles, Automated driving, IoT, Internet of Things, Europe, Socioeconomics, Intellectual property, Prototypes, data modeling, database, Intelligent vehicles, MongoDB, NodeJs, REST API, Restful API

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## Digital Labels: Influencing Consumers Trust and Raising Cybersecurity Awareness for Adopting Autonomous Vehicles

**Item Type** Conference Paper

**Author** W. Z. Khan

**Author** M. Khurram Khan

**Author** Q. -u. -A. Arshad

**Author** H. Malik

**Author** J. Almuhtadi

**Abstract** Autonomous vehicles (AVs) offer a wide range of promising benefits by reducing traffic accidents, environmental pollution, traffic congestion and land usage etc. However, to reap the intended benefits of AVs, it is inevitable that this technology should be trusted and accepted by the public. The consumer's substantial trust upon AVs will lead to its widespread adoption in the real-life. It is well understood that the preservation of strong security and privacy features influence a consumer's trust on a product in a positive manner. In this paper, we introduce a novel concept of digital labels for AVs to increase consumers awareness and trust regarding the security level of their vehicle. We present an architecture called Cybersecurity Box (CSBox) that leverages digital labels to display and inform consumers and passengers about cybersecurity status of the AV in use. The introduction of cybersecurity digital labels on the dashboard of AVs would attempt to increase the trust level of consumers and passengers on this promising

technology.

**Date** 10-12 Jan. 2021

**DOI** 10.1109/ICCE50685.2021.9427684

**Extra** Journal Abbreviation: 2021 IEEE International Conference on Consumer Electronics (ICCE)

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**Pages** 1-4

**Proceedings Title** 2021 IEEE International Conference on Consumer Electronics (ICCE)

**Conference Name** 2021 IEEE International Conference on Consumer Electronics (ICCE)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous Vehicles, Trust, Road safety, Reliability, Privacy, Technology Acceptance, Real-time systems, Computer security, Cybersecurity Awareness, Fasteners, Pollution, Security and Privacy

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# Driving Safety Risk Analysis and Assessment in a Mixed Driving Environment of Connected and Non-Connected Vehicles: A Systematic Survey

**Item Type** Journal Article

**Author** Z. Cheng

**Author** J. Zhu

**Author** Z. Feng

**Author** M. Yang

**Author** W. Zhang

**Author** J. Chen

**Abstract** With the continuous development of intelligent networks and autonomous driving technologies, heterogeneous traffic flow represented by conventional vehicles (CV), autonomous vehicles (AV), and connected and autonomous vehicles (CAV) have emerged, and consequently, the driving safety risk issues in this mixed driving environment have become increasingly complex. In a mixed and connected environment, different traffic streams in complex driving scenarios are intertwined with each other, and driving behaviours such as steering and lane-changing between various traffic streams are frequent, thus increasing the crash risk of vehicles. To understand the research methods, theoretical models, and system architectures in the field of driving safety analysis in this mixed-connected environment, this article reviews the driving safety risk progress from four major aspects: driving safety risk perception and identification, driving safety risk prediction, driving safety risk quantification, and driving safety risk early warning. By summarizing the existing research, it can be found that academics have achieved many achievements in urban driving safety risk evaluation in mixed-connected environments. Still, there are several problems and challenges that need to be solved, such as the stability and reliability of collaborative sensing systems of AV and CAV in complex traffic environments, the accuracy of object detection, the vehicle information security, the limitations of a single factor analysis used in driving safety risk assessment, and the accuracy of trajectory prediction for both the CV, AV, and CAV. By analyzing the limitations of the existing research, this article proposes a future research direction,

which provides a reference for the development of driving safety risk research.

**Date** May 2025

**DOI** 10.1109/TITS.2025.3526820

**Volume** 26

**Pages** 5747-5781

**Publication** IEEE Transactions on Intelligent Transportation Systems

**ISSN** 1558-0016

**Issue** 5

**Journal Abbr** IEEE Transactions on Intelligent Transportation Systems

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Safety, Accidents, Automobiles, Risk management, autonomous driving, Collision avoidance, driving safety risk early warning, driving safety risk perception and identification, driving safety risk prediction, driving safety risk quantification, Heterogeneous traffic flow, Collaboration, Real-time systems, Intelligent sensors, Analytical models, Information exchange

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## Driving the Future: Addressing Generational Trust and Ownership Barriers in the Adoption of Connected and Autonomous Vehicles

**Item Type** Conference Paper

**Author** C. Mutzenich

**Author** F. McVey

**Author** C. Martin

**Author** C. Harding

**Abstract** The advent of Connected and Autonomous Vehicles (CAVs) promises a transformative shift in transport, ushering in an era of shared mobility and interconnectedness. However, recent events highlight the challenges facing the widespread acceptance of CAVs and Mobility as a Service (MaaS). Drawing on insights from a survey of over 3,000 transport users in the UK, our study reveals two significant hurdles impeding the transition to shared autonomy. Firstly, user adoption presents a formidable challenge, slowing the pace of relinquishing control to driverless mobility, due to low trust and acceptance across generational groups. Secondly, the enduring preference for private vehicle ownership acts as a barrier to embracing shared mobility solutions. To address these challenges, we introduce the SASS Model, which categorises individuals into four distinct groups based on their emotional and rational inclinations towards CAVs and MaaS: Sceptics, Alarmists, Swing Voters, and Supporters. Each group necessitates a tailored approach to effectively communicate the benefits of CAVs. Strategies include addressing the concerns of Alarmists, providing reassurance to Sceptics, educating Swing Voters about the advantages of an automated future, and leveraging the backing of Supporters. Ultimately, the momentum of the automated (r)evolution will be driven by rational Sceptics and Swing Voters, highlighting the importance of targeted messaging and positioning by the automotive industry and transport providers to secure widespread adoption.

**Date**
2-5 June 2024

**DOI**
10.1109/IV55156.2024.10588833

**Extra**
Journal Abbreviation: 2024 IEEE Intelligent Vehicles Symposium (IV)

**ISBN**
2642-7214

**Pages**
397-402

**Proceedings Title**
2024 IEEE Intelligent Vehicles Symposium (IV)

**Conference Name**
2024 IEEE Intelligent Vehicles Symposium (IV)

**Date Added**
20/09/2026, 7:46:33 pm

**Modified**
20/09/2026, 7:46:33 pm

Tags:

Autonomous Vehicles, Trust, Acceptance, Education, Mobility as a service, Connected vehicles, Surveys, Industries, Buildings, SASS Model, Shared Mobility, Shared transport, User Adoption

Driving through the Concept Gridlock: Unraveling Explainability Bottlenecks in Automated Driving

**Item Type**
Conference Paper

**Author**
J. Echterhoff

**Author**
A. Yan

**Author**
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**Author**
A. Abdelraouf

**Author**
R. Gupta

**Author**
J. McAuley

**Abstract**

Concept bottleneck models have been successfully used for explainable machine learning by encoding information within the model with a set of human-defined concepts. In the context of human-assisted or autonomous driving, explainability models can help user acceptance and understanding of decisions made by the autonomous vehicle, which can be used to rationalize and explain driver or vehicle behavior. We propose a new approach using concept bottlenecks as visual features for control command predictions and explanations of user and vehicle behavior. We learn a human-understandable concept layer that we use to explain sequential driving scenes while learning vehicle control commands. This approach can then be used to determine whether a change in a preferred gap or steering commands from a human (or autonomous vehicle) is led by an external stimulus or change in preferences. We achieve competitive performance to latent visual features while gaining interpretability within our model setup. 1

**Date**
3-8 Jan. 2024

**DOI**
10.1109/WACV57701.2024.00718

**Extra**
Journal Abbreviation: 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

**ISBN**
2642-9381

**Pages**
7331-7340

**Proceedings Title**
2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

**Conference Name** 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Machine learning, Visualization, Uncertainty, Algorithms, Explainable, Autonomous Driving, accountable, Applications, ethical computer vision, fair, privacy-preserving, Predictive models, Linguistics, Process control, Transformers

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## Emerging Privacy and Trust Issues for Autonomous Vehicle Systems

**Item Type** Conference Paper

**Author** T. -H. Nguyen

**Author** T. G. Vu

**Author** H. -L. Tran

**Author** K. -S. Wong

**Abstract** In the awakening of cutting-edge technology, companies such as Apple, Waymo, and Tesla are racing to launch the industry's first fully autonomous car. Besides the technical challenges such as safety and infrastructure, privacy and data protection have attracted the autonomous vehicle industry and researchers' attention. In particular, it is hard for autonomous vehicle manufacturers to impose substantive privacy and security protections when different vendors and suppliers are involved in vehicle production. Although we know how much data autonomous vehicles will generate per day, there is a lack of knowledge of how the collected data will be used (e.g., real-time broadcasting and offline analytic). The privacy risks associated with data collection raise individual concerns in autonomous vehicle systems. For instance, when location information is combined with personal information, a person's details such as wealth status, profession, sexual association, and religion can be deduced. The misuse of present and historical travel patterns also puts someone susceptible to physical harm or stalking. Driven by mutual benefits or regulations, specific data must be shared in real-time or published for analysis or research purposes. This paper discusses the emerging privacy and trust issues that are essential to motivate the acceptance of autonomous vehicles operating on public roads.

**Date** 12-15 Jan. 2022

**DOI** 10.1109/ICOIN53446.2022.9687196

**Extra** Journal Abbreviation: 2022 International Conference on Information Networking (ICOIN)

**ISBN** 1976-7684

**Pages** 52-57

**Proceedings Title** 2022 International Conference on Information Networking (ICOIN)

**Conference Name** 2022 International Conference on Information Networking (ICOIN)

**Date Added** 20/09/2026, 7:46:32 pm

**Modified** 20/09/2026, 7:46:32 pm

**Tags:**

autonomous vehicles, self-driving car, Privacy, Regulation, privacy concerns, Roads, Data protection, Industries, Production, Real-time systems, trust issues

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## Enhanced Net Valence Model (NVM) for the Adoption of Autonomous Vehicles (AVs) by Novice Drivers

**Item Type** Conference Paper

**Author** A. A. A. Alshaafee

**Author** N. A. Iahad

**Abstract** Autonomous Vehicles (AVs, Level 4) can drive themselves from point A to point B without the need for any interaction from the driver because they are able to sense the surrounding environment. The few studies on adopting AVs tend to be narrowly focused on experienced drivers who have held a full license for a long time; there is no research yet on the adoption of AVs by novice drivers. Moreover, none of the previous studies has used the Net Valence Model (NVM) in investigating AV adoption. Realizing this gap, the aim of this study is to investigate factors influencing the adoption by novice drivers of Level 4 AVs. A conceptual model is proposed by extending NVM with three constructs as independent variables: personal innovativeness, alternatives, and social influence. Based on the proposed conceptual model, 16 hypotheses have been formulated which will be tested in the next phase of this research.

**Date** 2-3 Dec. 2019

**DOI** 10.1109/ICRIIS48246.2019.9073281

**Extra** Journal Abbreviation: 2019 6th International Conference on Research and Innovation in Information Systems (ICRIIS)

**ISBN** 2324-8157

**Pages** 1-6

**Proceedings Title** 2019 6th International Conference on Research and Innovation in Information Systems (ICRIIS)

**Conference Name** 2019 6th International Conference on Research and Innovation in Information Systems (ICRIIS)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

Autonomous vehicles, Autonomous Vehicles, Security, Artificial Intelligence, Social influence, Personal innovativeness, Psychology, Smart homes, Alternatives, Net Valence Model, Performance evaluation, Social network services

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## Enhancing Security and Privacy in Cyber-Physical Systems: Challenges and Solutions

**Item Type** Conference Paper

**Author** M. Sharma

**Abstract** Cyber-Physical Systems (CPS) represent a convergence of computational and physical processes, seamlessly integrating into sectors such as healthcare, transportation, and energy. As these systems become increasingly ubiquitous, the imperatives of security and privacy emerge as critical focal points, especially given the inherent vulnerabilities introduced by real-time operational constraints, device heterogeneity, and the multifaceted nature of physical-digital interactions. This paper aims to provide a comprehensive analysis of these challenges, bringing to the fore the issues of data sensitivity in applications such as healthcare, potential risks associated with user location tracking, and the looming threats of data aggregation and inference attacks. With these challenges as our backdrop, we journey into the realm of recent advancements, examining CPS-specific intrusion detection systems, the transformative potential of blockchain in ensuring data integrity, and the merits of communication protocols tailored for these systems. Furthermore, the discourse extends to the adoption of differential privacy techniques in CPS, the promise of homomorphic encryption in ensuring the sanctity of data processing, and user-centric mechanisms that afford individuals greater control over their privacy. As we encapsulate our findings, the research emphasizes the continuous evolution of threats and, concurrently, the countermeasures. In conclusion, the paper underscores the pressing need for a holistic, multi-disciplinary approach to security and privacy in CPS, urging the research community and industry stakeholders to champion collaborative efforts in fortifying these foundational systems.

**Date** 8-10 Jan. 2024

**DOI** 10.1109/CCWC60891.2024.10427691

**Extra** Journal Abbreviation: 2024 IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC)

**Pages** 0682-0686

**Proceedings Title** 2024 IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC)

**Conference Name** 2024 IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Security, Privacy, Transportation, Technological innovation, Cyber-physical systems, Real-time systems, Cyber-Physical Systems, Medical services

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## Ethical Issues in Autonomous Vehicle Programming- An Indian Perspective

**Item Type** Conference Paper

**Author** N. Nedunchezian

**Author** A. Anbuchejian

**Author** R. Subramanian

**Author** C. Arivazhagan

**Author** G. Chakkravarthy

**Abstract** This research paper analysis the ethical problems in programming an autonomous vehicle(AV) and suggests some guiding principles to be followed under Indian perspective. Unavoidable collision situations create moral dilemmas requiring consideration of the most acceptable and moral action of the vehicle. Many researchers proposed possible solutions for these dilemmas faced by the programmers in this domain. Each country have their own ethical principles and India is quit different in dealing with this problem. The moral values of each country and region differs and programming an autonomous vehicle also should follow the accepted ethical values of that country. For the acceptance of the AVs by public, a clear understanding of the cultural and ethical values must be the guiding principle in programming. Data security and privacy emerge as another key ethical considerations, requiring the need for added cyber security measures. The Indian traffic is one of the unique combination of many road users like animal driven carts, two wheeler, trucks, buses cars and multi axle vehicles along with pedestrians. Hence AV programming also requires an unique approach with combination of traffic rules, Indian ethical values, traffic climate and data security. The purpose of this research work is to advocate for transparency, accountability, and stakeholder engagement as fundamental principles for ensuring the ethical use of these technologies. The suggestion from this work gives valuable insight for policymakers, industry practitioners, and researchers in solving the complex ethical issues of autonomous vehicle technologies and will contribute in shaping the principles for the AVs programming in India.

**Date** 6-8 June 2025

**DOI** 10.1109/ETHICS65148.2025.11098210

**Extra** Journal Abbreviation: 2025 IEEE International Symposium on Ethics in Engineering, Science, and Technology (ETHICS)

**ISBN** 2996-3648

**Pages** 1-5

**Proceedings Title** 2025 IEEE International Symposium on Ethics in Engineering, Science, and Technology (ETHICS)

**Conference Name** 2025 IEEE International Symposium on Ethics in Engineering, Science, and Technology (ETHICS)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous vehicles, Safety, Autonomous Vehicle, Computer crime, Data security, ethics, Moral delima, programming, Programming, unavoidable collision, Stakeholders, Roads, Industries, Cultural differences, cyber security. AI, Ethics, Indian Ethical value

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## Examining Trust's Influence on Autonomous Vehicle Perceptions

**Item Type** Conference Paper

**Author** L. Tang

**Author** M. Bashir

**Abstract** The advent of autonomous vehicles (AVs) represents a transformative shift in transportation, promising to redefine mobility and alter our interaction with vehicles. Understanding public perceptions of AVs is crucial, as it influences the adoption and integration of this technology into society. This research conducts a comprehensive investigation into the factors that influence people's attitudes toward AVs, examining the associated benefits and concerns, as well as the extent of trust placed in this emerging technology. The primary objective is to gain a deeper understanding of the elements that contribute to human-machine trust in the context of AVs. The findings reveal a consistent pattern in the propensity to trust AVs and concerns regarding performance failures, both at individual and societal levels. From a societal perspective, enhanced locomotion independence is the primary benefit of AV deployment, contributing to increased accessibility and reduced reliance on conventional transportation systems. At the individual level, increased free time emerges as the foremost advantage. These findings provide AV developers and policymakers the critical insight when deploying autonomous vehicle systems.

**Date** 2-5 June 2024

**DOI** 10.1109/IV55156.2024.10588643

**Extra** Journal Abbreviation: 2024 IEEE Intelligent Vehicles Symposium (IV)

**ISBN** 2642-7214

**Pages** 667-673

**Proceedings Title** 2024 IEEE Intelligent Vehicles Symposium (IV)

**Conference Name** 2024 IEEE Intelligent Vehicles Symposium (IV)

**Date Added** 20/09/2026, 7:46:33 pm

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## Tags:

Ecosystems, Transportation, Regulation, Policy, Shape, Human factors, Ethics, and Regulations, Human Factors for Intelligent Vehicles, Human-machine systems

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## Explainable Artificial Intelligence for Autonomous Driving: A Comprehensive Overview and Field Guide for Future Research Directions

**Item Type** Journal Article

**Author** S. Atakishiyev

**Author** M. Salameh

**Author** H. Yao

**Author** R. Goebel

**Abstract** Autonomous driving has achieved significant milestones in research and development over the last two decades. There is increasing interest in the field as the deployment of autonomous vehicles (AVs) promises safer and more ecologically friendly transportation systems. With the rapid progress in computationally powerful artificial intelligence (AI) techniques, AVs can sense their environment with high precision, make safe real-time decisions, and operate reliably without human intervention. However, intelligent decision-making in such vehicles is not generally understandable by humans in the current state of the art, and such deficiency hinders this technology from being socially acceptable. Hence, aside from making safe real-time decisions,

AVs must also explain their AI-guided decision-making process in order to be regulatory-compliant across many jurisdictions. Our study sheds comprehensive light on the development of explainable artificial intelligence (XAI) approaches for AVs. In particular, we make the following contributions. First, we provide a thorough overview of the state-of-the-art and emerging approaches for XAI-based autonomous driving. We then propose a conceptual framework considering the essential elements for explainable end-to-end autonomous driving. Finally, we present XAI-based prospective directions and emerging paradigms for future directions that hold promise for enhancing transparency, trustworthiness, and societal acceptance of AVs.

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**DOI** 10.1109/ACCESS.2024.3431437

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### Tags:

Autonomous vehicles, Safety, Artificial intelligence, Accidents, safety, Autonomous driving, Regulation, Visualization, intelligent transportation systems, Explainable AI, Standards, explainable artificial intelligence, regulatory compliance

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## Explanations in Autonomous Driving: A Survey

**Item Type** Journal Article

**Author** D. Omeiza

**Author** H. Webb

**Author** M. Jirotko

**Author** L. Kunze

**Abstract** The automotive industry has witnessed an increasing level of development in the past decades; from manufacturing manually operated vehicles to manufacturing vehicles with a high level of automation. With the recent developments in Artificial Intelligence (AI), automotive companies now employ blackbox AI models to enable vehicles to perceive their environment and make driving decisions with little or no input from a human. With the hope to deploy autonomous vehicles (AV) on a commercial scale, the acceptance of AV by society becomes paramount and may largely depend on their degree of transparency, trustworthiness, and compliance with regulations. The assessment of the compliance of AVs to these acceptance requirements can be facilitated through the provision of explanations for AVs' behaviour. Explainability is therefore seen as an important requirement for AVs. AVs should be able to explain what they have 'seen', done, and might do in environments in which they operate. In this paper, we provide a comprehensive survey of the existing work in explainable autonomous driving. First, we open by providing a motivation for explanations by highlighting the importance of transparency, accountability, and trust in AVs; and examining existing regulations and standards

related to AVs. Second, we identify and categorise the different stakeholders involved in the development, use, and regulation of AVs and elicit their AV explanation requirements. Third, we provide a rigorous review of previous work on explanations for the different AV operations (i.e., perception, localisation, planning, vehicle control, and system management). Finally, we discuss pertinent challenges and provide recommendations including a conceptual framework for AV explainability. This survey aims to provide the fundamental knowledge required of researchers who are interested in explanation provisions in autonomous driving.

**Date** Aug. 2022

**DOI** 10.1109/TITS.2021.3122865

**Volume** 23

**Pages** 10142-10162

**Publication** IEEE Transactions on Intelligent Transportation Systems

**ISSN** 1558-0016

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### Tags:

autonomous vehicles, Autonomous vehicles, Artificial intelligence, Accidents, Automation, trust, Regulation, robotics, intelligent vehicles, human-machine interaction, explainable AI, regulations, standards, Stakeholders, Standards, accountability, Explanations

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## Exploring Ethical Issues and Challenges of Autonomous Vehicles in Different Countries and Cultures

**Item Type** Conference Paper

**Author** K. Shah

**Author** E. Guven

**Abstract** Autonomous vehicles (AVs) promise a revolutionary future of transportation, but ethical considerations pose significant challenges to widespread adoption. This paper delves into the intricate ethical challenges surrounding AVs, encompassing the perplexing "trolley problem" of unavoidable accidents, the thorny issue of liability attribution, the paramount concern of data privacy, the chilling prospect of hacking vulnerabilities, and the potential for widespread job displacement. We embark on a global exploration, examining how these ethical dilemmas manifest differently across the United States, Europe, Australia, China, and developing countries. By dissecting the ethical landscape in each region, we highlight the diverse cultural, legal, and societal contexts that influence the perception and management of these challenges. Each region offers a unique tapestry of approaches and solutions, from implementing robust regulatory frameworks and cultivating cutting-edge technological advancements to fostering community engagement and prioritizing educational initiatives. This analysis unveils the profound impact of cultural perspectives, highlighting how deeply ingrained values, such as individualism versus collectivism and risk tolerance, shape the understanding of ethical dilemmas and the formulation of solutions. Furthermore,

the paper investigates the possibility of creating a universally compliant self-driving car. We acknowledge the formidable challenges posed by the complex and challenging nature of international traffic laws, the heterogeneity of safety standards, and the disparities in infrastructure across the globe. The paper recognizes the crucial need for global collaboration and developing culturally sensitive solutions to ensure a future for autonomous vehicles grounded in safety, ethical principles, and responsible implementation. Through this comprehensive exploration, the paper strives to deepen our understanding of the moral dimensions of autonomous driving technology and ignite a global discourse on the responsible development and deployment of AVs.

**Date** 19-21 March 2025

**DOI** 10.1109/CISS64860.2025.10944682

**Extra** Journal Abbreviation: 2025 59th Annual Conference on Information Sciences and Systems (CISS)

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**Pages** 1-6

**Proceedings Title** 2025 59th Annual Conference on Information Sciences and Systems (CISS)

**Conference Name** 2025 59th Annual Conference on Information Sciences and Systems (CISS)

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### Tags:

Safety, Accidents, Transportation, autonomous, dilemma, ethical, self-driving, vehicles, Europe, Shape, Law, Cultural differences, Ethics, Standards, Global communication

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## Factors Affecting Trust in the Autonomous Vehicle: A Survey of Primary School Students and Parent Perceptions

**Item Type** Conference Paper

**Author** Y. Ma

**Author** S. Li

**Author** S. Qin

**Author** Y. Qi

**Abstract** With the rapid development of autonomous driving technology, autonomous vehicles are expected to perform all driving functions under certain conditions. In fact, some people have even proposed the concept of autonomous school buses. Accordingly, autonomous vehicles exhibit the great potential to become not only the primary alternative to human-driven vehicles during a child's journey to school, but also the vehicles used by today's children as they enter adulthood. However, the acceptance of such vehicles depends on both the children and their parents' trust in this new technology. Hence, this study explores the relationships among perceived usefulness, defects, and risks as well as negative emotions and degree of trust in autonomous vehicles. Based on an online survey of 131 primary school students and 133 parents, the current study concludes that the perceived benefits, perceived risks, and emotional responses influence people's trust in this new technology. Moreover, the results of a regression analysis find two critical factors that influence trust in autonomous vehicles, namely, perceived risk of

traffic safety and perceived defects in the vehicle. Additionally, while perceived usefulness is an exclusive predictor of adults' trust, negative emotions only significantly predict the trust of children in the vehicle. For trust in autonomous vehicles, these primary school students share similar perceptions with those of their parents. To our knowledge, the present study is the first to compare people's trust in autonomous vehicles between primary school students and their parents. These findings may inspire the design and promotion of autonomous vehicles for school journeys, thus resulting in increasingly more families trusting and accepting this new technology.

**Date** 29 Dec.-1 Jan. 2021

**DOI** 10.1109/TrustCom50675.2020.00277

**Extra** Journal Abbreviation: 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)

**ISBN** 2324-9013

**Pages** 2020-2027

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## Tags:

Autonomous vehicles, autonomous vehicle, Safety, Security, trust, Privacy, perceived risk, perceived usefulness, Regression analysis, emotion, Conferences, Emotional responses

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## Formal Methods for Connected Vehicle Protocols

**Item Type** Conference Paper

**Author** B. Stojanović

**Author** K. Hofer-Schmitz

**Abstract** A significant growth in connected vehicles industry arises new security issues. Security-by-design framework, including formal methods, has the potential to offer a higher security level in this domain. This paper provides an overview of the formal methods for connected vehicle protocols, including three most important formal verification fields in this domain, namely qualitative and quantitative analysis, functional correctness and security properties. It includes a systematic overview of formal methods categorized by these fields, used tools and considered protocols, including a short summary and open challenges for future work.

**Date** 26-27 Nov. 2019

**DOI** 10.1109/TELFOR48224.2019.8971034

**Extra** Journal Abbreviation: 2019 27th Telecommunications Forum (TELFOR)

**Pages** 1-4

**Proceedings Title** 2019 27th Telecommunications Forum (TELFOR)

**Conference Name** 2019 27th Telecommunications Forum (TELFOR)

**Date Added** 20/09/2026, 7:46:34 pm

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### Tags:

Security, security, Automotive engineering, Connected vehicles, formal verification, automotive industry, Industries, Protocols, Telecommunications, Statistical analysis, Delays, Systematics, connected cars, Formal methods, Formal verification, model checkers

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## Friday Forum

**Item Type** Conference Paper

**Abstract** The following topics are dealt with: MOSFET; silicon; elemental semiconductors; field effect transistors; SRAM chips; random-access storage; neural nets; learning (artificial intelligence); low-power electronics; CMOS memory circuits.

**Date** 9-14 June 2019

**DOI** 10.23919/VLSIT.2019.8776476

**Extra** Journal Abbreviation: 2019 Symposium on VLSI Technology

**ISBN** 2158-9682

**Pages** xvii-xvii

**Proceedings Title** 2019 Symposium on VLSI Technology

**Conference Name** 2019 Symposium on VLSI Technology

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### Tags:

Autonomous vehicles, Safety, Security, Automobiles, Heart, Automotive engineering, Sensors

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## Future on Wheels: Safeguarding Privacy in Tomorrow's Connected Vehicles-FUTURE-SP

**Item Type** Journal Article

**Author** F. Znidi

**Author** M. Morsy

**Author** H. Rathore

**Abstract** As connected and autonomous vehicles (CAVs) become increasingly integrated into daily transportation systems, they accumulate vast amounts of personal data, raising significant privacy concerns. This study focuses on navigating the balance between fostering technological innovation and ensuring robust privacy protections. It emphasizes the importance of encryption, transparent data management, and user-centric privacy settings to protect against unauthorized data access and surveillance. The analysis highlights the necessity for evolving legal frameworks that keep pace with rapid technological advancements in CAVs, advocating for adjustments to address emerging privacy challenges as they arise. Additionally, the study calls for international collaboration and legal adaptability to establish a regulatory

environment that supports the safe and privacy-conscious deployment of CAV technologies. By exploring public perceptions of CAV privacy and discussing advanced techniques such as differential privacy and federated learning, the study outlines effective strategies to enhance data security and foster trust among users. It stresses the importance of transparency and control over personal data to mitigate privacy risks and enhance public acceptance of CAVs. This comprehensive examination contributes to a multidisciplinary discourse, aiming to develop a cohesive global privacy framework that aligns CAV technological advancements with essential safety and privacy protections.

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### Tags:

Data privacy, Privacy, Data collection, autonomous driving, Connected vehicles, Vehicle-to-everything, data anonymization, encryption techniques, privacy concerns, vehicle-to-everything (V2X) communications, Real-time systems, Scalability, Encryption, Faces, Law, Protection

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## Highlighting the Future of Autonomous Vehicle Technology in 2020–2050

**Item Type** Conference Paper

**Author** N. B. Al Barghuthi

**Author** D. Pavithran

**Author** H. E. Said

**Abstract** As the world enters the age of the 4th industrial revolution, the future of work has re-emerged as a fundamental question, among policy-makers, business leaders, workers and the broader public around the globe. This is applied by accelerating innovations and adoption of ever more sophisticated automation technologies. Most of the existing research and reports have focused on the impact of automation in advanced industrial countries. In this paper, we explore the role of software engineering in autonomous transportation in 2050, its challenges and research trends. The exploration will be done through analysis of different reports and UAE government summit prediction. We intended to see the vision of the future of software engineering that prepares engineers for future emerging technology. This could contribute to the solution and design of international standards as well as increase the market confidence. In addition, this can also contribute towards a proper start-up of self-driving technology. We believe that a successful adoption of Autonomous Vehicles (AV) in developing a future automobile system will best be attained through an effective risk management strategy. As a result, the potential threat to the AV will further be investigated by industry and government entities to deal with the huge threats of cybersecurity

that might face the autonomous vehicles.

**Date** 28-29 Nov. 2018

**DOI** 10.1109/CTIT.2018.8649510

**Extra** Journal Abbreviation: 2018 Fifth HCT Information Technology Trends (ITT)

**Pages** 251-256

**Proceedings Title** 2018 Fifth HCT Information Technology Trends (ITT)

**Conference Name** 2018 Fifth HCT Information Technology Trends (ITT)

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## Tags:

autonomous vehicles, Autonomous vehicles, Automobiles, Transportation, Technological innovation, Market research, driverless, Faces, Computer security, Government, Standards, autonomous transportation, compliance, software engineering, Software engineering

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## How to Rationalise the Sampling of Test-Scenarios in Automated Driving Based on Criticality Metrics?

**Item Type** Conference Paper

**Author** H. Blache

**Author** P. -A. Laharotte

**Author** N. -E. E. Faouzi

**Abstract** The deployment of Automated and Connected Vehicles (ACV) into traffic requires certifications and validations guaranteeing high levels of safety, security and reliability. The underlying objective is to gain public acceptance by proving that automation systems might bring out a safer mobility. While plenty of methods to certify these systems are populating the literature, the scenario-based approach stands out by reducing the quantity of required Field tests to validate any new system at stake. In this study, we refine the scenario-based approach by proposing a proof of concept (PoC) for scenario reduction using criticality metrics. For this PoC, we weave a relationship between the a priori criticality of abstract functional scenarios and the words used to generate them. Once, the criticality of a subset of scenarios is qualified based on open field data (HighD), the Latent Dirichlet Allocation (LDA) clustering approach is used to generate topics and feature the relationship between observed criticality and semantics words applied to functional scenarios. The criticality degree of semantics words is used to predict the a priori criticality of unobserved functional scenarios.

**Date** 14-16 June 2023

**DOI** 10.1109/MT-ITS56129.2023.10241560

**Extra** Journal Abbreviation: 2023 8th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)

**Pages** 1-6

**Proceedings Title** 2023 8th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)

**Conference Name** 2023 8th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)

**Date Added** 20/09/2026, 7:46:33 pm**Modified** 20/09/2026, 7:46:33 pm**Tags:**

Safety, Security, Automation, Semantics, Connected vehicles, Connected and Automated Vehicle, critical scenarios, Latent Dirichlet Allocation, ontology, sampling strategy, Measurement, Resource management

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## Human Acceptance of Autonomous Mobility: A Literature Review

**Item Type** Conference Paper**Author** D. Hafiz

**Abstract** In recent years, with the emergence of autonomous mobility technologies and the multitude of services they offer, people's perception of Autonomous Mobility (AM) has surged. Individuals now have an array of choices at their disposal, enabling them to enhance the convenience and comfort of their trips. The trust people place in technology directly influences their embrace of such innovations. It is crucial to prioritize resources, policies, and activities focused on marketing, education, subsidization, and infrastructure advancement to foster greater public acceptance of autonomous mobility. Additionally, theories aimed at studying the factors that impact the acceptance of AM play a vital role. Promoting acceptance of autonomous mobility is vital for society, as attitudes towards use, perceived trust, and sense of gain play a significant role. Literature review and benchmarking existing experiences of autonomous technology is key in addressing public concerns and increasing awareness, ultimately influencing acceptability. The paper aims to provide a comprehensive understanding of the current state of knowledge in this field and identify future research opportunities to enhance people's acceptance of AM, especially in the Middle East and Gulf region.

**Date** 12-14 Feb. 2024**DOI** 10.1109/UVS59630.2024.10467165**Extra** Journal Abbreviation: 2024 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS)**Pages** 1-8**Proceedings Title** 2024 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS)**Conference Name** 2024 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS)**Date Added** 20/09/2026, 7:46:32 pm**Modified** 20/09/2026, 7:46:32 pm**Tags:**

Autonomous vehicles, public acceptance, Technological innovation, Education, autonomous mobility, Benchmark testing, Bibliographies, people's perception

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# Impact of AI Driven Predictive Privacy Mechanisms in UAV Integrated V2X 6G Networks

**Item Type** Conference Paper

**Author** N. K. Bairwa

**Author** G. Jain

**Author** S. Umrao

**Author** P. Ponde

**Abstract** The integration of unmanned aerial vehicles (UAVs) with vehicle-to-everything (V2X) communications in the forthcoming 6G networks presents unprecedented opportunities for connectivity and automation. However, this technological advancement also introduces significant privacy risks. In this work, privacy problems in V2X-6G networks with UAV integration are predicted and mitigated through the use of artificial intelligence (AI). By leveraging machine learning models, the research aims to develop predictive privacy mechanisms capable of detecting potential privacy breaches and autonomously adjusting privacy settings or alerting users in real-time. The methodology involves the creation and training of sophisticated machine learning algorithms for the purpose of analyzing large volumes of data that are transferred across V2X-6G networks. These models are designed to identify patterns indicative of privacy threats, such as unauthorized data access or information leakage, and respond proactively to protect user privacy. The study also explores the ethical considerations surrounding the deployment of AI in privacy management, addressing concerns related to AI biases, transparency, and user trust. This research contributes to the ongoing discourse on privacy protection in emerging communication networks and underscores the potential of AI in creating safer and more secure digital environments.

**Date** 20-21 Dec. 2024

**DOI** 10.1109/EmergIN63207.2024.10961785

**Extra** Journal Abbreviation: 2024 International Conference on Emerging Technologies and Innovation for Sustainability (EmergIN)

**Pages** 135-139

**Proceedings Title** 2024 International Conference on Emerging Technologies and Innovation for Sustainability (EmergIN)

**Conference Name** 2024 International Conference on Emerging Technologies and Innovation for Sustainability (EmergIN)

**Date Added** 20/09/2026, 7:46:34 pm

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## Tags:

Data privacy, Privacy, machine learning, Machine learning, Machine learning algorithms, Vehicle-to-everything, AI-driven privacy, predictive privacy mechanisms, UAV-integrated V2X-6G networks, Real-time systems, Ethics, Data models, 6G mobile communication, Autonomous aerial vehicles

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## Impact of autonomous vehicles on pedestrians' safety

**Item Type** Conference Paper

**Author** J. S. Brar

**Author** B. Caulfield

**Abstract** Autonomous vehicles (AVs) are currently being developed, and the subject is undergoing an intense transport study. They present enormous potential to enhance the safety, efficiency, and sustainability of road traffic. The development of autonomous vehicles (AVs) is rapidly moving forward with companies already performing or planning trials in public traffic environment. However, the aspect of how pedestrians will experience AVs and interact with them has largely been unexplored so far. The aim of this paper is to investigate the impact of autonomous vehicles on pedestrians' safety. An online survey was conducted which examined public opinion regarding Autonomous Vehicles. The survey yielded useable responses from 255 individuals. The major findings of these results is that the pedestrians showed high levels of concern about various issues regarding autonomous vehicles. They also felt positive about the autonomous vehicles and have optimistic expectations of the advantages of this technology.

**Date** 16-19 Oct. 2017

**DOI** 10.1109/ITSC.2017.8317963

**Extra** Journal Abbreviation: 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)

**ISBN** 2153-0017

**Pages** 714-719

**Proceedings Title** 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)

**Conference Name** 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)

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## Tags:

Autonomous vehicles, Autonomous Vehicles, Safety, Education, Roads, Conferences, Legged locomotion, pedestrian acceptance, Vehicle crash testing

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## Integrating AI With Cyber Security for Smart Industry 4.0 Application

**Item Type** Conference Paper

**Author** S. B. Goyal

**Author** A. S. Rajawat

**Author** R. K. Solanki

**Author** M. A. Majmi Zaaba

**Author** Z. A. Long

**Abstract** The advent of Smart Industry 4.0 has brought about significant advances in automation, efficiency, and productivity in industrial environments. However, these technological advances also pose new challenges for cybersecurity. The increasing interconnectivity of devices and systems within Smart Industry 4.0 environments makes them vulnerable to cyber attacks and security breaches. To address these challenges, there is a need to integrate AI with cybersecurity in Smart Industry 4.0 applications. AI can be used to analyze large amounts of data

from sensors and devices within the industrial environment to detect and prevent cyber threats. Machine learning algorithms can be trained to identify patterns of normal behavior and flag any deviations as potential security threats. There are different AI-based cybersecurity techniques that can be used, such as anomaly detection, intrusion detection, and predictive analytics. These techniques can be used in conjunction with other cybersecurity measures, such as firewalls, intrusion detection systems, and security information and event management (SIEM) tools, to provide a comprehensive approach to cybersecurity for Smart Industry 4.0 applications. Integrating AI with cybersecurity for Smart Industry 4.0 applications can help improve the security posture of industrial environments and reduce the risk of cyber attacks and data breaches. However, it is important to ensure that these technologies are properly implemented and configured to avoid unintended consequences and potential vulnerabilities. By carefully designing and implementing AI-based cybersecurity solutions, organizations can protect their industrial environments from cyber threats and ensure the reliability and safety of their operations.

**Date** 26-28 April 2023

**DOI** 10.1109/ICICT57646.2023.10134374

**Extra** Journal Abbreviation: 2023 International Conference on Inventive Computation Technologies (ICICT)

**ISBN** 2767-7788

**Pages** 1223-1232

**Proceedings Title** 2023 International Conference on Inventive Computation Technologies (ICICT)

**Conference Name** 2023 International Conference on Inventive Computation Technologies (ICICT)

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### Tags:

Safety, Artificial intelligence, Privacy, Intrusion detection, machine learning, Artificial Intelligence, Social networking (online), Deep Learning, Industry 4.0, Computer security, Cyber Security, Fourth Industrial Revolution

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## Integrating Cybersecurity in V2X: A Review of Simulation Environments

**Item Type** Journal Article

**Author** K. Herman Muraro Gularte

**Author** J. Paulo Javidi da Costa

**Author** J. A. R. Vargas

**Author** A. Santos da Silva

**Author** G. Almeida Santos

**Author** Y. Wang

**Author** C. Alfons Müller

**Author** C. Lipps

**Author** R. Timóteo de Sousa

**Author** W. de Britto Vidal Filho

**Author** P. Slusallek

**Author** H. Dieter Schotten

**Abstract** The deployment and acceptance of Vehicle-to-Everything (V2X) technologies are essential for the advancement of intelligent transportation systems, requiring robust simulation tools for development and validation. This paper offers an overview of leading V2X simulation tools, emphasizing their role in communication and cybersecurity applications. Existing surveys are reviewed, identifying gaps such as the limited coverage of less known tools such as CARLA and CarSim and the absence of detailed market analyses. A thorough assessment of the market significance of V2X simulation tools is provided, highlighting annual trends and profiling leading companies. Publication trends across major scientific databases are also examined, reflecting the growing interest and research diversity in V2X simulations. The review includes an analysis of crucial simulation tools such as SUMO, NS3, OMNeT++, Veins, and CARLA, emphasizing their contributions to V2X communications. Regarding cybersecurity, the study explores simulation parameters for mitigating security threats such as jamming, spoofing, Denial of Service (DoS), and eavesdropping. Furthermore, the increasing integration of different simulation tools is highlighted to address their individual limitations. Finally, emerging research opportunities are identified, advocating for the integration of co-simulation frameworks and AI-driven (Artificial Intelligence) approaches to improve the accuracy and resilience of V2X systems. This survey aims to guide future research efforts in developing secure, efficient, and scalable V2X communication technologies.

**Date** 2024

**DOI** 10.1109/ACCESS.2024.3504404

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## Tags:

Security, Artificial intelligence, Surveys, connected and automated vehicles, denial of service, eavesdropping, radio jamming, spoofing, trends in V2X simulations, V2X cybersecurity, V2X simulation tools, Vehicle-to-everything, Reviews, Cyberattack, Market research, Patents, Computer security, Testing

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## Interview Guide Development for Qualitative Research on Exploring Acceptance of Autonomous Vehicles in Thailand

**Item Type** Conference Paper

**Author** T. Sithanant

**Author** W. Chaiyasoonthorn

**Author** S. Chaveesuk

**Abstract** The aim of this article is to create a semi-structured interview guide for a qualitative research project centered on the exploring acceptance of autonomous vehicles (AVs) in Thailand. The widespread acceptance of AVs holds significant potential for enhancing the quality of life. While existing studies have predominantly focused on the acceptance of AVs through quantitative research and technology acceptance theories, this paper seeks to address this gap by proposing an innovative approach. This approach involves qualitative research rooted in the use of grounded theory methodology, employing semi-structured interviews to generate a novel theoretical framework. The collection of qualitative data will be facilitated through the adoption of semi-structured interviews, with the interview guide serving as the principal tool for data collection. The article will commence with a concise overview of AVs, pertinent technology acceptance theories, past research on AV acceptance, and the adoption of conventional car studies. Following that, a synthesis of knowledge and identification of gaps will be presented. Subsequently, a new research approach will be introduced, and a systematic procedure for crafting the semi-structured interview guide will be detailed and summarized.

**Date** 26-27 Oct. 2023

**DOI** 10.1109/ICITEE59582.2023.10317695

**Extra** Journal Abbreviation: 2023 15th International Conference on Information Technology and Electrical Engineering (ICITEE)

**ISBN** 2766-0419

**Pages** 340-346

**Proceedings Title** 2023 15th International Conference on Information Technology and Electrical Engineering (ICITEE)

**Conference Name** 2023 15th International Conference on Information Technology and Electrical Engineering (ICITEE)

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### Tags:

autonomous vehicle, Automobiles, acceptance, Data collection, interview guide, qualitative research, Interviews, Ethics, Data models, Electrical engineering, Systematics

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## Investigating ChatGPT on Reddit by Using Lexicon-Based Sentiment Analysis

**Item Type** Conference Paper

**Author** P. S. Putra

**Author** A. H. Lubis

**Author** R. Muliono

**Author** N. Khairina

**Author** Muhathir

**Author** M. R. Darmansyah

**Abstract** ChatGPT is one of the most favorable AI-driven applications, gaining widespread popularity and usage among users. Despite it provides numerous advantages, ChatGPT also introduces security vulnerabilities and ethical concerns, such as plagiarism, privacy issues, data security risks, and the potential for biased data

generation. Hence, identifying and addressing the sentiment of ChatGPT is essential, as it helps understand public perception and highlights areas requiring improvement. The research investigates the efficacy of various lexicon-based sentiment analysis tools, namely VADER, Liu-Hu, SentiWordNet, and AFINN, in capturing user sentiment from Reddit comments related to ChatGPT. The study emphasizes the importance of rigorous text preprocessing steps, including case folding, normalization, cleaning, tokenizing, stop-words removal, and stemming, to enhance the accuracy and reliability of sentiment analysis. The result highlights the presence of some consistency in identifying neutral sentiment across the tools, but also underscores the notable variability in detecting positive and negative sentiments. AFINN tends to identify a greater number of positive sentiments and a lesser number of negative ones, while VADER and SentiWordNet offer a more balanced and nuanced detection across all sentiment categories. Liu-Hu, meanwhile, occupies a middle ground with higher positive sentiment counts and lower negative sentiment counts. The evaluation of the sentiment clustering quality is conducted using the Silhouette Index, which measures that VADER and SentiWordNet exhibit higher scores (0.5326 and 0.5173, respectively) than other tools. The results of the sentiment analysis indicate that ChatGPT garners significantly more positive sentiment compared to negative and neutral sentiments.

**Date** 5-6 Sept. 2024

**DOI** 10.1109/ICITRI62858.2024.10699128

**Extra** Journal Abbreviation: 2024 International Conference on Information Technology Research and Innovation (ICITRI)

**Pages** 65-70

**Proceedings Title** 2024 International Conference on Information Technology Research and Innovation (ICITRI)

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**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Reliability, Sentiment analysis, Technological innovation, Social networking (online), Supervised learning, Chatbots, ChatGPT, Reddit, Sentiment Analysis, Accuracy, Indexes, Lexicon-Based, Natural Language Processing, Plagiarism, Tag clouds

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## Investigating the Efficacy of In-Process Feedback in Improving Human-Machine Interaction in Autonomous Vehicles

**Item Type** Conference Paper

**Author** J. C. Sekhar

**Author** J. Ramu

**Author** V. K. Pratap

**Abstract** As autonomous vehicles become more prevalent on our roads, the improvement of human-machine interaction (HMI) becomes increasingly critical. In-process feedback has emerged as a promising approach to enhancing HMI in autonomous

vehicles, but limited research exists on its effectiveness. The purpose of this study is to determine how effective in-process feedback is at enhancing HMI. To investigate the impact of in-process feedback on HMI, a within-subjects approach and a quasi-experimental design were used. Ninety participants with experience driving autonomous vehicles were recruited and divided into an investigational group receiving in-process feedback and a control group without such feedback. The study measured the efficacy of in-process feedback on HMIs through subjective and objective measures. The subjective measure utilized a questionnaire to assess participants' perceptions of their driving experience, including comfort, confidence, and trust in the autonomous vehicle. The objective measure analyzed participants' driving behavior, encompassing speed, lane keeping, and steering patterns. The experimental group outperformed the control group in terms of comfort, confidence, and trust in the driverless car, according to the results. The objective measure also indicated improved driving behavior in the investigational group, including smoother lane keeping and fewer instances of sudden braking and acceleration. In conclusion, this study supports the effectiveness of in-process feedback as an approach to enhancing HMI in autonomous vehicles. The findings have practical implications for the design and development of autonomous vehicle technology, ultimately contributing to the public's acceptance and adoption of this technology.

**Date** 29-30 July 2023

**DOI** 10.1109/AIC57670.2023.10263851

**Extra** Journal Abbreviation: 2023 IEEE World Conference on Applied Intelligence and Computing (AIC)

**Pages** 376-381

**Proceedings Title** 2023 IEEE World Conference on Applied Intelligence and Computing (AIC)

**Conference Name** 2023 IEEE World Conference on Applied Intelligence and Computing (AIC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Human computer interaction, Roads, Autonomous automobiles, Behavioral sciences, Accelerometers, Atmospheric measurements, Efficacy, Human-machine interaction (HMI), In-process feedback, Particle measurements, Quasi-experimental design

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## Is Greece Ready for Autonomous Vehicles? A Methodological Approach

**Item Type** Conference Paper

**Author** C. Mylonas

**Author** E. Mitsakis

**Author** A. Dolianitis

**Author** C. Chalkiadakis

**Abstract** Despite the debate regarding the timeframe and rate of penetration of Autonomous Vehicles, their potential benefits and implications have been widely recognized. Therefore, assessing the readiness of individual countries to adopt such technologies and adapt to their introduction is of particular importance. This paper aims to enrich our understanding of EU readiness regarding the

introduction of autonomous vehicle technologies by assessing the case of Greece. Thus, through a literature review, the criteria upon which such an assessment should be based are established and analyzed. Subsequently, the case of Greece is assessed based on those criteria by finding relevant sources that support and justify any assessment. Regardless of the outcome concerning the readiness of Greece, such an assessment should help identify areas in which focus should be given in order to ensure a smoother transition to such technologies. This contribution is expected to assist policy makers worldwide.

**Date** 20-23 Sept. 2020

**DOI** 10.1109/ITSC45102.2020.9294428

**Extra** Journal Abbreviation: 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)

**Pages** 1-6

**Proceedings Title** 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)

**Conference Name** 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous vehicles, Security, Data privacy, Automobiles, Roads, Certification, Guidelines

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## Legal considerations for the transportation of unaccompanied minors in autonomous vehicles

**Item Type** Conference Paper

**Author** S. Farmer

**Author** E. F. Judge

**Abstract** This article provides an overview of the variety of laws that currently regulate the transportation of children in motor vehicles in Canada, evaluates what changes are required if autonomous vehicles replace human drivers, and recommends the best options for legally regulating the transportation of unaccompanied children in autonomous vehicles.

**Date** 10-13 Nov. 2024

**DOI** 10.1109/WF-IoT62078.2024.10811331

**Extra** Journal Abbreviation: 2024 IEEE 10th World Forum on Internet of Things (WF-IoT)

**ISBN** 2768-1734

**Pages** 1-6

**Proceedings Title** 2024 IEEE 10th World Forum on Internet of Things (WF-IoT)

**Conference Name** 2024 IEEE 10th World Forum on Internet of Things (WF-IoT)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

autonomous vehicles, Autonomous vehicles, Safety, Vehicles, Automation, Automobiles, Technological innovation, Internet of Things, Surveillance, Law, Canadian law, children and the law, law and technology, motor vehicle law, Standards

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## Lightweight and Provably Secure Privacy-Preserving Implicit Authentication Protocol Using Weighted-MinHash for IoV Environment

**Item Type** Journal Article

**Author** H. Hu

**Author** C. Hsu

**Author** M. H. Au

**Author** J. Cui

**Author** L. Harn

**Author** Z. Zhao

**Abstract** Implicit authentication for the Internet of Vehicles (IoV) refers to the continuous and transparent identity verification of users by utilizing various vehicle intelligent terminals to collect the user's behavioral characteristics as authentication factors, ensuring that the connected vehicle remains under the control of a legitimate user at all times. Existing research on implicit authentication mainly focuses on enhancing the robustness and accuracy of the system, but ignores the privacy protection of user behavior characteristics. Since behavioral characteristics involve extremely sensitive personal information, it is imperative to design efficient and secure privacy protection protocols. However, there is currently little research that relies on cryptographic technology to ensure a high level of security while achieving efficient privacy protection. In this paper, we propose a provably secure privacy-preserving cryptographic protocol for implicit authentication in a semi-honest and non-colluding dual-cloud environment, where Weighted MinHash and Jaccard similarity are combined for the first time in an implicit authentication framework to achieve effective privacy protection. Weighted MinHash is used to process multimodal behavioral features, which not only solves the problem of behavioral feature exposure but also reduces the computational cost; while Jaccard similarity is used to evaluate the differences between features. The security of the proposed protocol is proven by cryptographic methods, i.e., using formal and informal security analysis as well as the formal security analysis tool AVISPA. Furthermore, the experimental verification and comparative analysis, e.g., the time complexity and communication bandwidth of connected vehicles, demonstrate that the proposed protocol is effective and practical. In contrast with the most advanced scheme, the time complexity and the communication bandwidth are decreased from  $O(n^3)$  to  $O(nm)(m \ll n)$  and from  $256qn^2 + 4096$  to  $768qm$ , respectively.

**Date** Oct. 2025

**DOI** 10.1109/TITS.2025.3605156

**Volume** 26

**Pages** 15394-15406

**Publication** IEEE Transactions on Intelligent Transportation Systems

**ISSN** 1558-0016

**Issue** 10

**Journal Abbr** IEEE Transactions on Intelligent Transportation Systems

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Security, Authentication, Privacy, connected vehicles, Connected vehicles, Cloud computing, Protocols, Protection, AVISPA, implicit authentication, Jaccard similarity, Privacy-preserving protocol, QR codes, Servers, Time complexity, Weighted MinHash

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## Media and Trust Influence Consumers' Acceptance of Self-driving Cars

**Item Type** Conference Paper

**Author** R. Du

**Author** H. Du

**Author** J. Wu

**Abstract** This study explores the public perception and acceptance of self-driving cars and its influencing factors. The acceptance model of new ternary interaction theory included self-efficacy and the external environment was innovatively proposed by constructing two different mass media (social media and traditional media). In our approach, we applied structural equation model and distribute questionnaires through a combination of online and offline methods with 437 valid response. Modeling result shows that the media has a significant impact on perceived value. Further, perceived security risk and perceived privacy risk significantly affect initial trust, while perceived value, self-efficacy and initial trust had positive effects on acceptance. Based on this learning, we suggest marketers paying more attention to the information release by mass media, promoting positive reports and improving word of mouth about driverless cars (self-driving cars) to increase consumer acceptance of self-driving cars.(Abstract)

**Date** 12-14 Aug. 2022

**DOI** 10.1109/ICTLE55577.2022.9901584

**Extra** Journal Abbreviation: 2022 10th International Conference on Traffic and Logistic Engineering (ICTLE)

**Pages** 57-61

**Proceedings Title** 2022 10th International Conference on Traffic and Logistic Engineering (ICTLE)

**Conference Name** 2022 10th International Conference on Traffic and Logistic Engineering (ICTLE)

**Date Added** 20/09/2026, 7:46:32 pm

**Modified** 20/09/2026, 7:46:32 pm

### Tags:

self-driving cars, Privacy, consumer acceptance, Social networking (online), structural equation model, intelligent transportation, Autonomous automobiles, Entertainment industry, Mathematical models, media, Media, Oral communication, social cognitive theory

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# Navigating the Future: Predictive Modeling of Safety Perceptions and Adoption Readiness of Autonomous Vehicles for Public Safety Enhancement

**Item Type** Conference Paper

**Author** S. Majumdar

**Author** A. Awasthi

**Author** S. E. Kirkley

**Abstract** The proliferation of autonomous vehicles (AVs) promises a shift towards more efficient and potentially safer roadways. However, the integration of AVs into existing transportation networks raises critical safety concerns, especially for vulnerable road users such as pedestrians and bicyclists. This paper examines the factors influencing these users' safety perceptions of sharing roadways with AVs. Leveraging survey data from Bike Pittsburgh and applying multinomial logit models, we assess how exposure to AVs, regulatory attitudes, and incidents involving AVs shape public trust and perceived safety. The study reveals that increased familiarity with AV technology correlates positively with safety perceptions, despite the historical impact of high-profile AV accidents. It also finds that public approval of AV testing is rising, suggesting a growing acceptance of AVs. Our analysis indicates a need for nuanced, evidence-based public policies that address the concerns of active transportation users and incorporate their perspectives into the formulation of safety guidelines in the era of AVs. Ultimately, the study highlights the importance of fostering positive interactions between AVs and vulnerable road users to promote harmonious coexistence and acceptance.

**Date** 6-8 Jan. 2025

**DOI** 10.1109/CCWC62904.2025.10903793

**Extra** Journal Abbreviation: 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)

**Pages** 00760-00768

**Proceedings Title** 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)

**Conference Name** 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous vehicles, Safety, Transportation, Surveys, Shape, Roads, Public security, Testing, Predictive models, adoption readiness, Autonomous Vehicles (AV), AV adoption, AV safety advancements, AV tech education, Computational modeling, logistic regression, multinomial logit model, predictive modeling, public safety, safety perceptions, technology familiarity

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## Neural networks as a motivation for humanity to evolve and survive

**Item Type** Conference Paper

**Author** N. Nosov

**Author** S. M. V. Lomonosov

**Abstract** the article discusses the directions and prospects for the development of such a direction as neural networks, as well as the consequences of their implementation for daily use.

**Date** 18-19 Aug. 2021

**DOI** 10.1109/EnT52731.2021.00030

**Extra** Journal Abbreviation: 2021 International Conference Engineering Technologies and Computer Science (EnT)

**Pages** 130-132

**Proceedings Title** 2021 International Conference Engineering Technologies and Computer Science (EnT)

**Conference Name** 2021 International Conference Engineering Technologies and Computer Science (EnT)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Security, self-driving cars, security, Neural networks, Autonomous automobiles, Computer science, neural networks

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## Parallel Surfaces: Service-Oriented V2X Communications for Autonomous Vehicles

**Item Type** Journal Article

**Author** S. Han

**Author** F. -Y. Wang

**Author** G. Luo

**Author** L. Li

**Author** F. Qu

**Abstract** The application of reconfigurable intelligent surfacess (RIS) in the Internet of Vehicles (IoV) is rapidly progressing, with a focus on intelligent signal manipulation, adaptive channel estimation, and real-time reconfiguration to optimize communication performance. While RIS technology holds significant potential for improving connectivity in IoV and vehicle settings, its fixed configuration falls short in meeting the dynamic demands in IoV. To address the diverse requirements of IoV services, including stringent safety norms, societal acceptance, and cost efficiency, this article proposes a parallel surfaces system to facilitate service-oriented V2X communications for autonomous vehicles. By harnessing artificial models of the physical RIS-assisted IoV system, conducting extensive computational experiments, and parallel execution, the proposed system enables real-time monitoring, predictive analysis, and optimization. Parallel intelligence empowers efficient and adaptable channel estimation, configuration, and energy management, thereby establishing a more intelligent and secure RIS-assisted IoV infrastructure. In addition, the parallel surfaces system encompasses descriptive, predictive, and prescriptive intelligence in IoV for addressing coverage gaps, enabling hotspot offloading, facilitating multiple antenna signal transmission, and enhancing high-precision positioning.

**Date** Nov. 2023

**DOI** 10.1109/TIV.2023.3324507  
**Volume** 8  
**Pages** 4536-4545  
**Publication** IEEE Transactions on Intelligent Vehicles  
**ISSN** 2379-8904  
**Issue** 11  
**Journal Abbr** IEEE Transactions on Intelligent Vehicles  
**Date Added** 20/09/2026, 7:46:33 pm  
**Modified** 20/09/2026, 7:46:33 pm

### Tags:

autonomous vehicles, Autonomous vehicles, V2X, Reliability, Vehicle-to-everything, Vehicle dynamics, Real-time systems, Channel estimation, Parallel intelligence, reconfigurable intelligent surfaces, Wireless communication

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## Perception and technology adoption trends for autonomous vehicles: Educational case study

**Item Type** Conference Paper

**Author** A. Mushtaq

**Author** S. Riaz

**Author** H. Mohd

**Author** A. Saleh

**Abstract** In recent years, self-driving cars have caused substantial attention and discussion. The effects of such autonomous cars on current load of traffic in UAE is a widespread topic in the research community. The literature gathered empirical evidences on adoption patterns of self-driving vehicles, people's likely use of them, and how that might influence amount of travel, mode choice, auto ownership, and other travel behavior decisions. However, autonomous vehicles are not yet in the market, therefore, there is a need to investigate travelers' behavior in terms of their perception concerns associated with autonomous vehicles and adoption trends. In this research paper, local contexts and opinions were gathered and examined. Questionnaire was designed and distributed to 100 different participants. The study investigates perception concerns and their association with innovation adoption trends. Theory of Diffusion of Innovation has been used for cross-comparison and classification in Trend-Analysis. One-way Analysis of Variance, Post-hoc Multiple Comparison, and Correlation Coefficients using Scatter-dot Plot techniques were deployed to test Hypotheses. The data was collected and analyzed using SPSS v.20. This research concludes that safety is biggest perception concern of travelers, while the adoption trend more or less follows the diffusion pattern. However, there is a need to educate people by introducing simulators in the market to reduce Chasm Transition phase, which has been found to be marginally significant in adoption analysis of autonomous vehicles.

**Date** 6 Feb.-5 April 2018

**DOI** 10.1109/ICASET.2018.8376923

|                          |                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Extra</b>             | Journal Abbreviation: 2018 Advances in Science and Engineering Technology International Conferences (ASET) |
| <b>Pages</b>             | 1-5                                                                                                        |
| <b>Proceedings Title</b> | 2018 Advances in Science and Engineering Technology International Conferences (ASET)                       |
| <b>Conference Name</b>   | 2018 Advances in Science and Engineering Technology International Conferences (ASET)                       |
| <b>Date Added</b>        | 20/09/2026, 7:46:33 pm                                                                                     |
| <b>Modified</b>          | 20/09/2026, 7:46:33 pm                                                                                     |

**Tags:**

Autonomous vehicles, Autonomous Vehicles, Safety, Automobiles, Privacy, Market research, Autonomous automobiles, Adoption Trends, User Perception

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## Performance evaluation of Autonomous and Connected Electric Vehicles (ACEVs) towards buying intention in India: Forward trend in energy efficiency

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Item Type</b>         | Conference Paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Author</b>            | N. Vikas Babu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Author</b>            | M. Damle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Abstract</b>          | Performance evaluation of autonomous and connected electric vehicles (ACEVs) with consumers' buying behaviour will assist the consumer community towards decision making. The views of EV owners as unconventional, consumer behaviour and perception of the next-generation vehicular revolution analysed towards adopting electric vehicles as it is in transformative stages in India. This study endeavours to better understand consumers' buying intentions toward ACEVs. The methodology is analysis with descriptive Statistics and exploratory factor analysis using the data collected through primary sources with data collection formulated with a review of secondary research concerning the Indian Automobile Industry. The analysis predicts consumer attitudes toward automated vehicles. The findings render the understanding of the purchase intention against the performance evaluation basis recognised theories of consumer behaviour with emphasis on objectivity in decision-making and the rationale for satisfaction within the performance parameter. The implications are for the industry forming the perception of willing consumers with an evaluation of parameters of acceptance toward these adoptions of electric and autonomous vehicles. The study outcomes give insights which are quintessential to businesses with determinants related to the adoption of this revolution which can benefit businesses, policymakers and marketers in building their strategies.. |
| <b>Date</b>              | 18-19 Nov. 2022                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>DOI</b>               | 10.1109/IIHC55949.2022.10060349                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Extra</b>             | Journal Abbreviation: 2022 International Interdisciplinary Humanitarian Conference for Sustainability (IIHC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Pages</b>             | 74-79                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Proceedings Title</b> | 2022 International Interdisciplinary Humanitarian Conference for Sustainability (IIHC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

**Conference Name** 2022 International Interdisciplinary Humanitarian Conference for Sustainability (IIHC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Electric vehicles, Decision making, Consumer behavior, Costs, Uncertainty, Industries, Performance evaluation, Au-tonomous, Automobile, Buying Intention, Consumer, Consumer Behaviour, Electric Vehicles

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## PhaDe: Practical Phantom Spoofing Attack Detection for Autonomous Vehicles

**Item Type** Journal Article

**Author** F. Lin

**Author** H. Yan

**Author** J. Li

**Author** Z. Liu

**Author** L. Lu

**Author** Z. Ba

**Author** K. Ren

**Abstract** Despite their prevalence and indispensability in the perception modules of autonomous vehicles, cameras have shown susceptibility to numerous attacks. Among them, the phantom spoofing attack is of significant concern. In such attacks, malefactors employ electronic display devices like projectors and display monitors to generate deceptive objects, thereby duping the object detectors in autonomous vehicles. However, existing detection methodologies are narrowly focused on a single device category, ignoring the multitude of devices that could be leveraged for attacks. Furthermore, the artificial modality-based solution presently in use lacks efficacious fusion mechanisms. In response to these limitations, we propose PhaDe, a practical deep learning-based system adept at detecting phantom spoofing attacks from a variety of and even unfamiliar attack devices. Our approach introduces two image processing techniques to construct artificial modalities and further advances a multi-head self-attention MSA-based fusion module for more versatile integration of disparate modalities. To boost the generalization capacity of our system against novel, unseen attacks, we incorporate two representation-level losses to align feature distributions from various domains. Evaluations conducted on our own dataset, encompassing fake objects from several device types, attest to the efficacy of our system. Our results indicate an accuracy of 98.80% on familiar domains and a detection success rate of 94.03% on unfamiliar domains. Additionally, PhaDe demonstrates a swift response time, fulfilling the practicality requisites.

**Date** 2024

**DOI** 10.1109/TIFS.2024.3376192

**Volume** 19

**Pages** 4199-4214

**Publication** IEEE Transactions on Information Forensics and Security

**ISSN** 1556-6021

**Journal Abbr** IEEE Transactions on Information Forensics and Security

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous vehicles, Object detection, Cameras, deep learning, Performance evaluation, detection system, Monitoring, multi-modal fusion, phantom attack, Phantoms, Training, Vehicles security

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## Privacy on Autopilot: Exploring User Attitudes Toward Data Use in Autonomous Vehicles

**Item Type** Conference Paper

**Author** N. Z. AlRais

**Author** S. M. AlHajeri

**Author** H. Almansoori

**Author** A. AlAwadhi

**Author** A. Panthakkan

**Author** S. A. Amin

**Abstract** The rapid evolution of autonomous vehicle (AV) technology driven by artificial intelligence (AI) and big data analytics is transforming the landscape of modern transportation. Yet, this technological progress brings pressing concerns about data privacy and user trust, particularly due to the constant collection, sharing, and analysis of real-time data. This study explores user perceptions of data privacy in autonomous vehicles, with a focus on the context of Dubai. Utilizing machine learning (ML) techniques and k-fold crossvalidation, we analyzed user data across various demographic factors, including age, gender, and education, to accurately predict privacy concerns. Among several classifiers K-Nearest Neighbors, Decision Tree, Support Vector Machine (SVM), Random Forest, Neural Networks, and AdaBoost Gradient Boosting demonstrated the highest predictive accuracy. The results underscore the importance of robust data governance frameworks, public awareness initiatives, and cross-sector collaboration among policymakers, AI researchers, and the automotive industry. Future work should consider longitudinal studies, the application of advanced deep learning models, and the development of comprehensive regulatory strategies to foster a privacy-conscious, AI-enabled mobility ecosystem.

**Date** 15-18 Sept. 2025

**DOI** 10.1109/ASET66891.2025.11427971

**Extra** Journal Abbreviation: 2025 Advances in Science and Engineering Technology International Conferences (ASET)

**ISBN** 2831-6878

**Pages** 1-5

**Proceedings Title** 2025 Advances in Science and Engineering Technology International Conferences (ASET)

**Conference Name** 2025 Advances in Science and Engineering Technology International Conferences (ASET)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Data privacy, Artificial intelligence, Support vector machines, Transportation, Smart mobility, Neural networks, Predictive modeling, Real-time systems, Predictive models, Pressing, Random forests, Space vector modulation

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## Promoting User Acceptance in Autonomous Driving

**Item Type** Conference Paper

**Author** W. Titov

**Author** T. Schlegel

**Abstract** The User Acceptance in Autonomous Vehicle is the main subject that has received the attention of researcher and professional in all around the world. The main objective of the paper is to encourage people towards technological acceptance of Autonomous vehicles. All benefit, after technological difficulties, do not come without a certain amount of challenges. The present challenges of Autonomous Vehicle are Assurance of system and Software, Sensing and Connectivity, Judgment, and Verification and Validation. This paper helps to provide more information regarding this technology. It helps the users to understand how safe and best this technology to use. To better predict, explain and increase User acceptance, we need to better understand why people will not accept or reject this technology. To use the knowledge obtained in order to foresee improved knowledge has on the actions of both the automotive industry and government. These papers provide practical evidence of road accidents as well as it compares Manual and Automatic Cars, which helps users to decide what good for them. There are questions asked by every individual who has to be answered. Among those questions, there is a common question asked by every user "How safe are you in an autonomous vehicle"?

**Date** 5-8 July 2022

**DOI** 10.23919/SpliTech55088.2022.9854288

**Extra** Journal Abbreviation: 2022 7th International Conference on Smart and Sustainable Technologies (SpliTech)

**Pages** 1-6

**Proceedings Title** 2022 7th International Conference on Smart and Sustainable Technologies (SpliTech)

**Conference Name** 2022 7th International Conference on Smart and Sustainable Technologies (SpliTech)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Safety, Regulation, User Acceptance, Sensors, Manuals, Software, Road accidents, Autonomous Technology, Government, Road Safety, User Behavior

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## Public Acceptance to utilize Autonomous Vehicles AV's in the United Arab Emirates, An Empirical Study

**Item Type** Conference Paper

**Author** K. Hamarsheh

**Abstract** The new era of the Autonomous Vehicles AV technology is an essential phenomenon that shapes the Smart Cities transformation in the near future. Autonomous Vehicles AV's (AV for short) provides new concept of mobility solutions to the public that are yet to arrive soon in the middle east countries. United Arab Emirates UAE being one of the leading countries in middle east in terms of AVs tests is seriously taking steps towards the implementation of AV in Public Transportation, nevertheless, very few studies have taken serious research on how the ordinary public are ready to accept and use the AV Transportation. This paper analyzes the main influential factors that shape the acceptance and perception of the public to use AV transportation particularly in UAE. A total of twelve factors extracted from Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), in addition to Behavioral and Non-behavioral Theories were screened out. Some of these factors were found to significantly affect the AV acceptance and intention to use. This paper focused on Technology Acceptance, Technology Trust, Anxiety, Perceived Safety, Reliability, Performance Expectancy, Privacy, Security, Social Influence, Psychological Factors, Efforts Expectancy, and Legal Liability as some of the main influential factors that shape the acceptance and perception of the Public to use AV in UAE.

**Date** 4-6 Nov. 2024

**DOI** 10.1109/ICTMOD63116.2024.10959127

**Extra** Journal Abbreviation: 2024 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)

**ISBN** 2159-5119

**Pages** 1-6

**Proceedings Title** 2024 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)

**Conference Name** 2024 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Autonomous Vehicles, Safety, Security, Reliability, Privacy, TAM, Public Acceptance, Psychology, Technology Acceptance, Structural Equation Modelling, UTAUT, Shape, Mathematical models, Law, Anxiety, Anxiety disorders, AV, CFA, Confirmatory Factor Analysis, Efforts Expectancy, Legal Liability, Perceived Safety, Performance Expectancy, Psychological Factors, SEM, Social Influence, Technology management, Technology Trust

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## Public Accountability: Understanding Sentiments towards Artificial Intelligence across Dispositional Identities

**Item Type** Conference Paper

**Author** B. Richardson

**Author** D. Prioleau

**Author** K. Alikhademi

**Author** J. E. Gilbert

**Abstract** Artificial Intelligence (AI) and Machine Learning (ML) have been influential across many industries. Companies, nearly every day, are finding new means and methods of benefiting from these technologies. Despite this prevalence, individuals still report a significant level of distrust towards AI and its applications. To rehabilitate the relationship between AI and its consumers, developers must expose these new technologies to consumers and include them in the process of critiquing and assisting in the improvement of such technologies. The goal of this work is to introduce a new initiative towards an Ethical AI society. Participants are given the opportunity to learn about modern applications of AI and the space to reflect on these technologies. It is found that across the exemplified technologies, differences of opinions are significantly correlated to specific dispositional identities, such as gender and computing experience. Furthermore, trends of trust across the general public are compared to that of students enrolled in a computer science course. These results depict vastly differing opinions across technologies which validate the need for public exposure and critique. This work highlights the need for researchers and developers to investigate opinions across dispositional identities, including race, gender, socioeconomic status, etc. The study has shown to be beneficial, with over 70% of individuals reporting having learned about a new application of AI.

**Date** 12-15 Nov. 2020

**DOI** 10.1109/ISTAS50296.2020.9462184

**Extra** Journal Abbreviation: 2020 IEEE International Symposium on Technology and Society (ISTAS)

**ISBN** 2158-3412

**Pages** 489-496

**Proceedings Title** 2020 IEEE International Symposium on Technology and Society (ISTAS)

**Conference Name** 2020 IEEE International Symposium on Technology and Society (ISTAS)

**Date Added** 20/09/2026, 7:46:34 pm

**Modified** 20/09/2026, 7:46:34 pm

### Tags:

algorithmic bias, Machine learning, ethics, Market research, Industries, ML, Computer science, AI, Companies, Electric breakdown, fairness, Modems, trust in AI

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## Public attitudes towards autonomous mini buses operating in real conditions in a Hellenic city

**Item Type** Conference Paper

**Author** E. Portouli  
**Author** G. Karaseitanidis  
**Author** P. Lytrivis  
**Author** A. Amditis  
**Author** O. Raptis  
**Author** C. Karaberi

**Abstract** Public attitudes are a key factor shaping the demand and market for autonomous vehicles. Previous studies are limited to a priori attitudes since such vehicles are not yet fully accessible to the public. A survey among 200 passengers of autonomous mini buses in real operation in the city of Trikala shows that they were well accepted and integrated in the people's everyday life and that there were no major concerns as regards safety and security on board. A survey involving 519 adult Trikala citizens revealed that regular users of the autonomous mini bus were younger people, experienced with driving automation systems, who would prefer fully autonomous than manual driving and who would use an autonomous vehicle. Although the findings should be verified with bigger samples and in more cities and traffic environments, they are encouraging towards the introduction of autonomous buses in urban environments.

**Date** 11-14 June 2017

**DOI** 10.1109/IVS.2017.7995779

**Extra** Journal Abbreviation: 2017 IEEE Intelligent Vehicles Symposium (IV)

**Pages** 571-576

**Proceedings Title** 2017 IEEE Intelligent Vehicles Symposium (IV)

**Conference Name** 2017 IEEE Intelligent Vehicles Symposium (IV)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Safety, Security, Urban areas, Europe, Interviews, Roads

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## Public Perception of Autonomous and Shared Autonomous Vehicles: Case Study of Amman

**Item Type** Conference Paper

**Author** O. Alharasees

**Author** H. A. Al Hasanat

**Author** D. Alothman

**Abstract** This study evaluates the public perception and adoption potential of autonomous vehicles (AVs) and shared autonomous vehicles (SAVs) in Amman, Jordan. Key factors influencing public perception and adoption include safety concerns, infrastructure readiness, and regulatory frameworks. The research highlights the necessity of governmental support and well-structured regulations. Despite recent advancements in Amman's transportation sector, challenges such as congestion, inadequate public transport, and infrastructure maintenance persist. Significant steps, including enhancing community preparedness and addressing public concerns, are crucial for successful AV and SAV integration. The study offers insights and recommendations to guide policymakers and stakeholders in

fostering Amman's sustainable urban mobility ecosystem. Collaborative efforts, pilot programs, and targeted outreach are essential for advancing A V and SA V adoption.

**Date** 12-13 Sept. 2024

**DOI** 10.1109/ICTIIA61827.2024.10761581

**Extra** Journal Abbreviation: 2024 2nd International Conference on Technology Innovation and Its Applications (ICTIIA)

**Pages** 1-5

**Proceedings Title** 2024 2nd International Conference on Technology Innovation and Its Applications (ICTIIA)

**Conference Name** 2024 2nd International Conference on Technology Innovation and Its Applications (ICTIIA)

**Date Added** 20/09/2026, 7:46:32 pm

**Modified** 20/09/2026, 7:46:32 pm

### Tags:

Autonomous vehicles, Safety, Urban mobility, Transportation, Public perception, Regulatory frameworks, Regulation, Technological innovation, Autonomous vehicles (AVs), Urban areas, Stakeholders, Amman transportation, Collaboration, Infrastructure readiness, Market research, Shared autonomous vehicles (SAVs), Socioeconomics

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## Public Perception of Connected and Automated Vehicles: Benefits, Concerns, and Barriers from an Australian Perspective

**Item Type** Journal Article

**Author** A. Matin

**Author** H. Dia

**Abstract** This study investigates the attitudes and concerns of the Australian public toward connected and autonomous vehicles (CAVs), and the factors influencing their willingness to adopt this technology. Through a comprehensive survey, a diverse group of respondents provided valuable insights toward various CAV scenarios such as riding in a vehicle with no driver, self-driving public transport, self-driving taxis, and heavy vehicles without drivers. The results highlight the significant impact of safety concerns about automated vehicles on individuals' attitudes across all scenarios. Higher levels of concern were associated with more negative attitudes, and a strong correlation between concerns and opposition underlines the necessity of addressing these apprehensions to build public trust and promote CAV adoption. Interestingly, nearly 70% of respondents felt uncomfortable driving next to a CAV, but they displayed more confidence in adopting automated public transport in the near future. Additionally, around 40% of participants indicated a strong willingness to purchase a CAV, primarily driven by the desire to reduce their carbon footprint and safety considerations. Notably, respondents with health conditions or disability exhibited heightened interest (almost double those without health conditions) in CAV technology. Gender differences emerged in attitudes and preferences toward CAVs, with women expressing a greater level of concern and perceiving higher barriers to CAV deployment. This emphasizes the importance of employing targeted approaches to address the specific concerns of different demographics. The study also

underscores the role of trust in technology as a significant barrier to CAV deployment, ranking high among respondents' concerns. To overcome these challenges and facilitate successful CAV deployment, various strategies are suggested, including live demonstrations, dedicated routes for automated public transport, adoption incentives, and addressing liability concerns. The findings from this study offer valuable insights for government agencies, vehicle manufacturers, and stakeholders in promoting the successful implementation of CAVs. By understanding societal acceptance and addressing concerns, decision-makers can devise effective interventions and policies to ensure the safe and widespread adoption of CAVs in Australia. Moreover, vehicle manufacturers can leverage these results to consider design aspects that align with passenger preferences, thereby facilitating the broader acceptance and adoption of CAVs in the future. Finally, this research provides a significant contribution to the understanding of public perception and acceptance of CAVs in the Australian context. By guiding decision-making and informing strategies, the study lays the foundation for a safer and more effective integration of CAVs into the country's transportation landscape.

**Date** June 2024

**DOI** 10.26599/JICV.2023.9210028

**Volume** 7

**Pages** 108-128

**Publication** Journal of Intelligent and Connected Vehicles

**ISSN** 2399-9802

**Issue** 2

**Journal Abbr** Journal of Intelligent and Connected Vehicles

**Date Added** 20/09/2026, 7:46:32 pm

**Modified** 20/09/2026, 7:46:32 pm

### Tags:

autonomous vehicles, Safety, Vehicles, road safety, Automation, Automobiles, public acceptance, Australia, Urban areas, Surveys, connected and automated vehicles (CAVs), stakeholder consultations

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## Public perception of V2X-technology - evaluation of general advantages, disadvantages and reasons for data sharing with connected vehicles

**Item Type** Conference Paper

**Author** T. Schmidt

**Author** R. Philipsen

**Author** P. Themann

**Author** M. Ziefle

**Abstract** This work aims at an evaluation of vehicle-to-infrastructure (V2X)-technology through the users' perspective. The technical opportunities of connected vehicles are affected by the acceptance of the technology and possible draw-backs on the privacy and data-security side. With a three-tiered research approach, this work identified beforehand argument lines in focus group discussions, which enabled a quantitative approach to evaluate positively and negatively perceived features of V2X-technology. Also gender related differences can be displayed. Further, the results of the second quantitative study indicate that although users who already

have experience with driver assistance systems are more willing to share (personal) data to use V2X-technology, the overall sample is very reserved with respect to sharing driver-related data. Future research on user diversity and cultural differences is outlined.

**Date** 19-22 June 2016

**DOI** 10.1109/IVS.2016.7535565

**Extra** Journal Abbreviation: 2016 IEEE Intelligent Vehicles Symposium (IV)

**Pages** 1344-1349

**Proceedings Title** 2016 IEEE Intelligent Vehicles Symposium (IV)

**Conference Name** 2016 IEEE Intelligent Vehicles Symposium (IV)

**Date Added** 20/09/2026, 7:46:32 pm

**Modified** 20/09/2026, 7:46:32 pm

## Tags:

Safety, Data privacy, Vehicles, Privacy, Connected vehicles, Data security

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## Public Perception Toward Tesla Fatal Accidents: Neural Network Approach

**Item Type** Conference Paper

**Author** N. Sohaee

**Abstract** This study endeavors to comprehensively evaluate the impact that fatal accidents involving autonomous vehicles, with a focus on incidents associated with Tesla, have on the public's perception of this technology. To achieve this, we systematically collect and examine user interactions on Twitter, encompassing the period both before and after such accidents occur. By employing a sophisticated Siamese neural network model for our analysis, we are able to effectively identify and quantify shifts in public sentiment and discourse regarding autonomous vehicles. The results of our investigation reveal a nuanced landscape of public opinion, characterized by a significant alteration in perception following these incidents. Specifically, our findings delineate a complex dynamic where, alongside an increased acceptance of autonomous vehicle technology, there exists a persistent and substantial concern regarding its safety and reliability. This study not only sheds light on the immediate reaction of the public to fatal accidents involving autonomous vehicles but also contributes to a deeper understanding of the broader societal implications of adopting such technology. Through this analysis, we underscore the importance of addressing public safety concerns and enhancing communication strategies to foster a more informed and balanced discourse around the future of autonomous vehicle technology.

**Date** 1-5 July 2024

**DOI** 10.1109/QRS-C63300.2024.00144

**Extra** Journal Abbreviation: 2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C)

**ISBN** 2693-9371

**Pages** 1091-1096

**Proceedings Title** 2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C)

**Conference Name** 2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C)  
**Date Added** 20/09/2026, 7:46:32 pm  
**Modified** 20/09/2026, 7:46:32 pm

### Tags:

Autonomous vehicles, Security, Accidents, Social networking (online), Neural networks, Software reliability, Blogs, fatal accidents, Public security, siamese network, Software quality, tesla, twitter, Vehicle dynamics

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## Public Perceptions Regarding Data Privacy in Use of Smart Speaker

**Item Type** Conference Paper

**Author** F. Gabriel

**Author** J. Cornelis

**Author** G. S. Handaja

**Author** A. Gui

**Abstract** This study aims to determine public perceptions of smart speakers regarding data privacy. Digital technology such as artificial intelligent (AI) are growing every year and implemented in many devices, which includes smart speakers. Smart speaker as virtual assistant makes people can interact with system using internet of things. There are several issues related with this, one of them is data privacy because smart speakers track user voice input and processed it through the internet. This research uses quantitative methods through online questionnaires and the population are Indonesia citizens that have not yet use smart speakers with the sample size of 187 samples. A modified Technology Acceptance Model (TAM), is used for this research. which consists variable factors such as perceived privacy (PP), perceived enjoyment (PE), perceived usefulness (PU), perceived ease of use (PEOU), attitudes (AT), behavior intention (BI), and perceived trust (PT). This research revealed that perceived trust does not influence on attitude toward using smart speaker, but perceived usefulness, perceived ease of use, perceived privacy, and perceived enjoyment positively influence the attitude towards smart speakers. Perceived usefulness and attitude positively also influence the respondent intention in using smart speakers. The limitation of this study is small sample size and limited survey area.

**Date** 17-19 Dec. 2024

**DOI** 10.1109/ICICYTA64807.2024.10913028

**Extra** Journal Abbreviation: 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)

**Pages** 796-801

**Proceedings Title** 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)

**Conference Name** 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

Security, Data privacy, Artificial intelligence, Privacy, User experience, data privacy, Regulation, Technology acceptance model, Internet of Things, Surveys, internet of things, Virtual assistants, artificial intelligent, smart speaker, virtual assistant

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## Readiness, Safety, and Privacy on Adopting Autonomous Vehicle Technology: UAE Case Study

**Item Type** Conference Paper

**Author** N. B. Al Barghuthi

**Author** H. Said

**Abstract** Different perspectives discuss autonomous vehicle (AV) as predicted future transportation that includes design, engineering, economics, computer science, social, economic, legislation, and ethical aspects. On the other hand, AV technology offers new problems such as new engineering aspects, social and ethical problems, and presents in the form of an idealized decision-making problem. There are many benefits to AV vehicles, which are at the forefront of technology development. However, the level of cybersecurity provided by this technology is poorly understood. Cyber-attacks against AV can have severe implications for safety, privacy, manufacturers' reputation, and overall perception. In this research, we examine public opinion regarding the implications and challenges that might face AV technology in the United Arab Emirates (UAE). We claim that the successful adoption of AV in future transport systems will be achieved by reasonable risk management approaches that deal with different potential threats of cybersecurity. This survey produced usable responses from 204 participants. The significant result was that the respondents had not previously heard of vehicle-related technology; however, most had a positive first opinion on technology. Around 85% of participants felt that the expected benefits were likely to occur. They expressed a high level of concern regarding security and performance issues. Respondents stated that safety is the most crucial aspect of driverless vehicles. Mostly, it stated that personal communication devices should integrate with the AV's internet connection. The primary outcomes are that the population feels positive about AV and have optimistic expectations regarding its benefits. While maintaining some concerns, most of the people are willing to drive the AV.

**Date** 20-21 Nov. 2019

**DOI** 10.1109/ITT48889.2019.9075090

**Extra** Journal Abbreviation: 2019 Sixth HCT Information Technology Trends (ITT)

**Pages** 47-52

**Proceedings Title** 2019 Sixth HCT Information Technology Trends (ITT)

**Conference Name** 2019 Sixth HCT Information Technology Trends (ITT)

**Date Added** 20/09/2026, 7:46:32 pm

**Modified** 20/09/2026, 7:46:32 pm

**Tags:**

autonomous vehicles, Safety, Security, cybersecurity, Accidents, Automobiles, privacy, self-driving, ethics, Roads, driverless, Face, Licenses, social-aspects

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## Risks and Ethical Concerns in Cyber Security with Advancements of Artificial Intelligence - A Systematic Review

**Item Type** Conference Paper

**Author** A. Perez

**Author** S. Anwar

**Abstract** Contribution: This study offers an analysis of cybersecurity for continuing advancements in artificial intelligence (AI). Specifically, we synthesized the literature and highlighted the possible cybersecurity risks and ethical concerns associated with advancements in AI. Background: Recent advancements in AI have implicated various risks and ethical concerns regarding cybersecurity. Prior literature studies have highlighted a plethora of these issues. We have conducted a systematic review to analyze the issues presented in the prior literature. Analyzing the issues presented has allowed us to answer one research question. Research Question: Our study answered one research question: Given the continuous advancements in artificial intelligence, what are cybersecurity's potential risks and ethical concerns? Methodology: We utilized the IEEE Xplore database to answer the research question. After setting the search criteria to journal articles posted between 2019 and 2024, we found 67 articles relevant to our keywords. After screening, we finalized 16 articles to address our research question. The articles selected were analyzed using qualitative thematic analysis. Articles were then categorized into three different themes: (1) social engineering, (2) misinformation and fake news, and (3) hacking. Findings: Our findings indicate that researchers have informed us about the malicious use of AI in cybersecurity by classifying attacks into three themes. These themes indicate the manner of attack. Furthermore, we find that the major risks and ethical concerns regarding the application of AI in cybersecurity relate to the preservation of privacy. As the field of artificial intelligence continues to grow, we see that it is imperative to continue monitoring the safety of this technology. For researchers to serve the public, future work should aim to translate technical terminology into more digestible phrases for those not knowledgeable in the field.

**Date** 13-16 Oct. 2024

**DOI** 10.1109/FIE61694.2024.10893551

**Extra** Journal Abbreviation: 2024 IEEE Frontiers in Education Conference (FIE)

**ISBN** 2377-634X

**Pages** 1-5

**Proceedings Title** 2024 IEEE Frontiers in Education Conference (FIE)

**Conference Name** 2024 IEEE Frontiers in Education Conference (FIE)

**Date Added** 20/09/2026, 7:46:34 pm

**Modified** 20/09/2026, 7:46:34 pm

**Tags:**

Safety, Artificial intelligence, artificial intelligence, privacy, Systematic literature review, Computer crime, Large language models, ethical concerns, Computer security, Ethics, cyber attacks, cybersecurity risk, Deepfakes, Terminology, Translation

# Safeguarding the V2X Pathways: Exploring the Cybersecurity Landscape Through Systematic Review

Item Type

Journal Article

Author

K. Herman Muraro Gulate

Author

J. Alfredo Ruiz Vargas

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J. Paulo Javidi da Costa

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A. Santos da Silva

Author

G. Almeida Santos

Author

Y. Wang

Author

C. Alfons Müller

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C. Lipps

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R. Timóteo de Sousa Júnior

Author

W. de Britto Vidal Filho

Author

P. Slusallek

Author

H. Dieter Schotten

Abstract

Vehicle-to-Everything (V2X) communication, essential for enhancing road safety, driving efficiency, and traffic management, must be robust against cybersecurity threats for successful deployment and acceptance. This survey comprehensively explores V2X security challenges, focusing on prevalent cybersecurity threats such as jamming, spoofing, Distributed Denial of Service (DDoS), and eavesdropping attacks. These threats were selected due to their prevalence and ability to compromise the integrity and reliability of V2X systems. Jamming can disrupt communications, spoofing can lead to data and identity manipulation, DDoS attacks can saturate system resources, and eavesdropping can compromise user privacy and information confidentiality. Addressing these major threats ensures that V2X systems are robust and secure for successful deployment and widespread acceptance. This work makes significant contributions to the field of V2X cybersecurity, starting with a thorough review and categorization of existing survey papers, providing a clear map of the current research landscape, and identifying areas needing further study. An extensive review uncovered a global landscape of V2X cybersecurity research. We highlight contributions from the leading countries in scientific publications and patent innovations, with notable advancements from leading corporations. This work educates and informs on the current state of V2X cybersecurity and identifies emerging trends and future research directions based on a year-by-year analysis of the literature and patents. The findings underscore the evolving cybersecurity landscape in V2X systems and the importance of continued innovation and research in this critical field. The survey navigates the complexities of securing V2X communications, emphasizing the necessity for advanced security protocols and technologies, and highlights innovative approaches within the global scientific and patent research context. By providing a panoramic view of the field, this survey sets the stage for future advancements in V2X cybersecurity.

Date

2024

**DOI** 10.1109/ACCESS.2024.3402946  
**Volume** 12  
**Pages** 72871-72895  
**Publication** IEEE Access  
**ISSN** 2169-3536  
**Journal Abbr** IEEE Access  
**Date Added** 20/09/2026, 7:46:32 pm  
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### Tags:

Autonomous vehicles, Safety, Security, Jamming, Connected vehicles, Surveys, Eavesdropping, Vehicle-to-everything, Cyberattack, connected and automated vehicle, cyberattacks and attacks, intellectual property in V2X cybersecurity, V2X security, INDEX TERMS Vehicle-to-everything, Intellectual property, Patents

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## Security Analysis of Connected Autonomous Vehicles (CAVs): Challenges, Issues, Defenses, and Open Problems

**Item Type** Conference Paper  
**Author** N. M. I. Chowdhury  
**Author** R. Hasan  
**Abstract** Our world is evolving into a landscape of smart cities, with connected autonomous vehicles (CAVs) occupying a central role in the paradigm of smart mobility. To ensure road and public safety, transportation productivity, employment opportunities, and cost-effectiveness, the CAVs can be the pioneers in the overall transportation system. Renowned automobile companies are transforming and expanding their market into CAVs as future cars by creating additional AV wings in their industries. However, having modern technologies, these cars face unique challenges in operation and can be intended to be attacked by malicious entities. In this survey paper, we classified the attacks in terms of issues arising into three sectors of autonomous vehicles (AVs) – AV forensics, AV communications, and OTA updates for AV. We emphasize the security analysis by discussing several challenges, attack scenarios and defenses, and a few open problems. We anticipate that our paper will serve as a valuable resource for readers and apprentices working on attacks and defenses on connected autonomous vehicles.  
**Date** 14-15 May 2024  
**DOI** 10.1109/WFPST58552.2024.00036  
**Extra** Journal Abbreviation: 2024 IEEE World Forum on Public Safety Technology (WFPST)  
**Pages** 81-86  
**Proceedings Title** 2024 IEEE World Forum on Public Safety Technology (WFPST)  
**Conference Name** 2024 IEEE World Forum on Public Safety Technology (WFPST)  
**Date Added** 20/09/2026, 7:46:33 pm  
**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

safety, Transportation, security, communication, Smart cities, Surveys, CAVs, Roads, Public security, Industries, forensics, OTA Update, Productivity

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## Security and Privacy of Artificial Intelligence with Ethical Concerns

**Item Type** Conference Paper

**Author** Y. Li

**Abstract** In recent years, artificial intelligence has entered the public eye across various fields, enhancing productivity and increasing social welfare. However, it has also raised significant security and privacy concerns due to the ethical uncertainties associated with this technology, requiring researchers to establish and refine the rules. This paper aims to categorize the security and privacy issues brought by AI and their relation to social ethics, analyze this complex topic from technical, social, and legal perspectives, and propose the current optimal solutions while predicting future developments.

**Date** 23-26 Aug. 2024

**DOI** 10.1109/DSC63484.2024.00098

**Extra** Journal Abbreviation: 2024 IEEE 9th International Conference on Data Science in Cyberspace (DSC)

**Pages** 660-667

**Proceedings Title** 2024 IEEE 9th International Conference on Data Science in Cyberspace (DSC)

**Conference Name** 2024 IEEE 9th International Conference on Data Science in Cyberspace (DSC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

**Tags:**

Security, Artificial intelligence, Privacy, privacy, security, Regulation, Technological innovation, Uncertainty, Law, Ethics, Productivity, artificial intelligence (ai), data protection, ethical challenges, Transforms

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## Self-Driving Cars: An Epitome of Technological Innovation

**Item Type** Conference Paper

**Author** K. Barman

**Author** A. Mondal

**Author** R. Dutta

**Author** N. S. Kalita

**Author** P. S. Chatterjee

**Abstract** Autonomous vehicles (AVs) are bringing a massive revolution in leveraging advancements in the fields of sensor technologies, artificial intelligence and machine learning algorithms. This paper focuses on providing extensive details on technical foundations, ethical concerns, legal challenges, and societal

implications. Key technologies including perception systems, decisionmaking algorithms and experimental frameworks are analysed to understand their role in enabling safe and efficient autonomy. Global regulatory discrepancies, privacy problems, and ethical quandaries in accident scenarios are also examined. The difficulties experienced by systems such as Waymo's autonomous taxis and Tesla's Full-Self Driving Beta are demonstrated by real-world case studies. The results highlight the significance of filling key research gaps in the areas of societal repercussions, legal uniformity, and scalability. The future of autonomous vehicles is expected to be inclusive, safe, and morally upright through the promotion of interdisciplinary cooperation. It also considers the economic implications of widespread AV adoption and particularly the impact on employment and infrastructure transformation when making efforts to address these challenges. Exploration of the integration of AVs in existing urban structure which has emphasized the need for adaptive traffic management and energy-efficient solutions have also been done. Trust and acceptance must also be built by raising public awareness and educating about the capabilities and limitations of AVs.

**Date** 8-9 Feb. 2025

**DOI** 10.1109/ESIC64052.2025.10962726

**Extra** Journal Abbreviation: 2025 International Conference on Emerging Systems and Intelligent Computing (ESIC)

**Pages** 754-758

**Proceedings Title** 2025 International Conference on Emerging Systems and Intelligent Computing (ESIC)

**Conference Name** 2025 International Conference on Emerging Systems and Intelligent Computing (ESIC)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Autonomous vehicles, Artificial intelligence, artificial intelligence, Privacy, Technological innovation, Public transportation, decision-making algorithms, ethical quandaries, experimental frameworks, global regulatory discrepancies, interdisciplinary cooperation, legal uniformity, machine learning algorithms, Machine learning algorithms, perception systems, privacy problems, sensor technologies, societal repercussions, Systematic literature review, Scalability, Law, Ethics, Standards

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## Self-Driving Vehicles: Exploring Preferred Adoption Scenarios - A Preliminary Study

**Item Type** Conference Paper

**Author** M. Khoyudia

**Author** L. Iskandar

**Author** M. Mariani

**Abstract** This initial research study examines the preferences of the general public concerning the adoption of various scenarios associated with self-driving vehicles (SDV). This study aimed to evaluate the readiness of one hundred volunteers to

employ self-driving vehicles (SDVs) in fifteen scenarios. The simulations included highway driving with options for human intervention and autonomous public transportation systems. We conducted descriptive and cluster analyses to identify significant patterns and classify comparable scenarios. Participants exhibited a moderate to high desire to employ autonomous vehicles in different scenarios. People expressed a strong willingness to use these technologies in settings such as universities and airports and for daily life. However, there was an apparent hesitation regarding the acceptance of autonomous vehicles for long trips and the concept of self-driving taxis operating without any human supervision. The findings reveal four distinct preference clusters: 'Highway Focus' (highway driving with manual control option), 'Safety Conscious' (hesitant toward full automation), 'Convenience Seekers' (specific situations such as traffic congestion), and 'Enthusiastic Adopters' (high overall acceptance). These clusters show various attitudes toward adopting self-driving vehicles (SDVs) and the importance of developing tailored strategies to tackle specific challenges and improve overall acceptance. There is a general openness to autonomous vehicles, particularly in specific scenarios or regulated settings. However, safety concerns and the need for control still pose major challenges to widespread use. Public acceptance will likely take time and depend on how well these issues are addressed. Further research and development are needed to improve safety, reliability, and public trust in driverless technology.

**Date** 17-19 Dec. 2024

**DOI** 10.1109/ICICYTA64807.2024.10912905

**Extra** Journal Abbreviation: 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)

**Pages** 1101-1105

**Proceedings Title** 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)

**Conference Name** 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Safety, Automation, Reliability, self-driving cars, Ecosystems, Research and development, technology adoption, Traffic congestion, Public transportation, Stakeholders, Autonomous automobiles, Human factors, adoption scenario

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## Software Transparency as a Key Requirement for Self-Driving Cars

**Item Type** Conference Paper

**Author** L. M. Cysneiros

**Author** M. Raffi

**Author** J. C. Sampaio do Prado Leite

**Abstract** Self-Driving cars is a fast-growing area of study both in academia and industry. It is part of a broader domain which involves the development of software for Highly Automated Vehicles (HAV) and notions extracted from Artificial Intelligence/Autonomous Systems (AI/AS). There are many challenges that must

be overcome to deliver self-driving cars in a manner that is readily accepted by consumers and society. Studies have shown that although many people are comfortable with the idea of AI helping them to operate their houses or schedule appointments, not many people are comfortable with the idea of cars being driven by AI algorithms. At the same time, insurance companies are concerned about vehicle liability issues and how to demonstrate who/what caused an accident. We believe that self-driving cars that demonstrate transparency in their operations will increase consumer trust which is pivotal to its acceptance and will pave the way for its commercialization and daily use. In this work, we investigate how to pursue the elicitation and modeling of transparency as a Non-Functional Requirement (NFR) to produce self-driving cars that are more robust.

**Date** 20-24 Aug. 2018

**DOI** 10.1109/RE.2018.00-21

**Extra** Journal Abbreviation: 2018 IEEE 26th International Requirements Engineering Conference (RE)

**ISBN** 2332-6441

**Pages** 382-387

**Proceedings Title** 2018 IEEE 26th International Requirements Engineering Conference (RE)

**Conference Name** 2018 IEEE 26th International Requirements Engineering Conference (RE)

**Date Added** 20/09/2026, 7:46:33 pm

**Modified** 20/09/2026, 7:46:33 pm

## Tags:

Autonomous vehicles, Automobiles, Autonomous automobiles, Industries, Autonomous Systems, Non-Functional Requirements, Highly Automated Vehicles, and Transparency, Software

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## Strategic Leadership in Autonomous Vehicle Integration for Enhanced Urban Mobility

**Item Type** Conference Paper

**Author** K. Sudharson

**Author** K. Selvi

**Author** C. T. C

**Author** L. Anitha

**Author** A. SureshKumar

**Author** V. Subhashini

**Abstract** The introduction of autonomous vehicles (AVs) in urban mobility represents an opportunity to revolutionize land use, transport efficiency, safety and environmental sustainability. This paper investigates critical strategic leadership approaches optimized to achieve successful AVs deployment on city streets using modern technologies, based on machine learning algorithms fused with perception sensors, V2X communication and agent-based traffic simulation. Using case studies of the top smart cities, this paper identifies best practices and strategic methodologies which have led to effective AV implementation in these areas, particularly in terms of congestion reduction, land use planning and sustainability. Collaboration with public stakeholders and proactive leadership for sustainable transport not only reduce traffic and emissions, but also foster

economic development and social equity. Such insights emphasize the capacity of AVs to contribute to more human-centric, enduring metropolises.

|                          |                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Date</b>              | 25-26 July 2025                                                                               |
| <b>DOI</b>               | 10.1109/ICCD564403.2025.11209717                                                              |
| <b>Extra</b>             | Journal Abbreviation: 2025 2nd International Conference on Computing and Data Science (ICCD5) |
| <b>Pages</b>             | 1-5                                                                                           |
| <b>Proceedings Title</b> | 2025 2nd International Conference on Computing and Data Science (ICCD5)                       |
| <b>Conference Name</b>   | 2025 2nd International Conference on Computing and Data Science (ICCD5)                       |
| <b>Date Added</b>        | 20/09/2026, 7:46:32 pm                                                                        |
| <b>Modified</b>          | 20/09/2026, 7:46:32 pm                                                                        |

**Tags:**

autonomous vehicles, Autonomous vehicles, Safety, Sensor fusion, machine learning, Sustainable development, Traffic control, smart cities, Machine learning algorithms, Smart cities, Vehicle-to-everything, traffic management, Stakeholders, Leadership, vehicle-to-everything

Subtype Divergences of Trust in Autonomous Vehicles: Towards Optimisation of Driver–Vehicle Trust Management

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Item Type</b> | Conference Paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Author</b>    | M. S. Seet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Author</b>    | A. Dragomir                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Author</b>    | I. Mathialagan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Author</b>    | L. Y. Ann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Author</b>    | Z. b. Zaid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Author</b>    | S. L. Ramapatna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Author</b>    | N. V. Thakor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Author</b>    | A. Bezerianos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Abstract</b>  | Trust determines public acceptance and uptake of autonomous vehicles (AV). Against popular assumption, trustin-automation is not a unitary construct, but comprises trust subtypes that have different behavioural properties and implications. Understanding trust subtypes-specifically competence-based trust (CT) and integrity-based trust (IT)-is crucial for improving public communication about AVs, analysing trustdependent driver behaviours and designing trust-recovering interfaces. However, these issues have been overlooked in most past research. As a pioneering step, the goal of this research was to analyse how experience with AV failures affect CT and IT. After experience with AV driving errors, both trust subtypes were reduced, with CT showing greater reduction. Structural equation modelling revealed CT to be the primary contributor to acceptance for driving automation, with stronger subsequent impact on preference for fully autonomous (SAE L5) than on semi-autonomous driving (SAE L3). These findings inform that trust-repairing interface should target CT after driving errors, especially for higher automation levels where humans are further removed from the loop. Future directions concerning CT-IT interactions, and the impact of AV anthropomorphic design and connected vehicle cyber-security on IT are |

discussed.

**Date** 20-23 Sept. 2020

**DOI** 10.1109/ITSC45102.2020.9294495

**Extra** Journal Abbreviation: 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)

**Pages** 1-6

**Proceedings Title** 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)

**Conference Name** 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)

**Date Added** 20/09/2026, 7:46:33 pm

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## Tags:

Autonomous vehicles, Autonomous Vehicles, Vehicles, Automation, Reliability, Human Factors, Roads, Behavioural Modelling, Junctions, Mathematical model, Trust in Automation

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## Survey on cyber security of CAV

**Item Type** Conference Paper

**Author** Q. He

**Author** X. Meng

**Author** R. Qu

**Abstract** With the ever fast developments of technologies in science and engineering, it is believed that CAV (connected and autonomous vehicles) will come into our daily life soon. CAV could be used in many different aspects in our lives such as public transportation and agriculture, and so on. Although CAV will bring huge benefits to our lives and society, issues such as cyber security threats, which may reveal drivers' private information or even pose threat to driver's life, present significant challenges before CAV can be utilised in our society. In computer science, there is a clear category of cyber security attacks while there is no specific survey on cyber security of CAV. This paper overviews different passive and active cyber security attacks which may be faced by CAV. We also present solutions of each of these attacks based on the current state-of-the-art, and discuss future improvements in research on CAV cyber security.

**Date** 19-21 May 2017

**DOI** 10.1109/CPGPS.2017.8075153

**Extra** Journal Abbreviation: 2017 Forum on Cooperative Positioning and Service (CPGPS)

**Pages** 351-354

**Proceedings Title** 2017 Forum on Cooperative Positioning and Service (CPGPS)

**Conference Name** 2017 Forum on Cooperative Positioning and Service (CPGPS)

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## Tags:

CAV, Automobiles, Authentication, Autonomous Driving, Encryption, Cyber Security, Global Positioning System, Receivers

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## The Role of Sensors in Autonomous Vehicles and its Impact on Smart Transportation Systems

**Item Type** Conference Paper

**Author** A. Wunnava

**Author** H. Kalra

**Author** A. Pattanaik

**Author** S. Lalithakumari

**Author** R. K. Yadav

**Author** L. Khanna

**Abstract** Once at the forefront of a transformation in the transportation industry, autonomous vehicles have all but disappeared from our collective discourse. Most autonomous vehicles depend on sensors to detect what is around and make some decisions without human interference. Sensors are paramount in functioning AVs since they give real-time information regarding the vehicle's location, rate, and surrounding objects. The sensors here are for all purposes; some use cameras, while others use lidar, radar, and several ultrasonic sensors. Combining both gives the vehicle a 360-degree view and helps it maneuver safely through even the most complex environments. AV-cum-smart transport systems can help improve traffic flow and road safety and increase overall transportation efficiency. Sensors are critical to harnessing these benefits and allow the AVs to work by way of other vehicles or organization, such as circulation signals and road ciphers. This communication gets dynamic route planning so that the time of travel and fuel can be saved. Nevertheless, several challenges remain to be addressed with sensor reliability under any conditions; however adverse, they may still face serious opposition and cybersecurity. Further research and development in the area of sensor technology are imperative to bring AVs into common use, as well as for using smart transportation systems effectively.

**Date** 4-6 March 2025

**DOI** 10.1109/AUTOCOM64127.2025.10956194

**Extra** Journal Abbreviation: 2025 International Conference on Automation and Computation (AUTOCOM)

**Pages** 431-435

**Proceedings Title** 2025 International Conference on Automation and Computation (AUTOCOM)

**Conference Name** 2025 International Conference on Automation and Computation (AUTOCOM)

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### Tags:

Autonomous vehicles, Autonomous Vehicles, Cybersecurity, Road safety, Reliability, Smart transportation, Cameras, Sensors, Transportation Efficiency, Transportation industry, Vehicle dynamics, Computer security, Intelligent sensors, Sensor systems

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## The Unforeseen in Unmanned Vehicles

**Item Type** Conference Paper

**Author** P. Maxwell

**Author** M. Nowatkowski

**Abstract** The development of unmanned vehicle technology is rapidly proceeding and will result in numerous advances in autonomous vehicles. Most of the research effort to date focuses on the safe and effective operation of these vehicles that will allow them to integrate into society. A research gap exists though in the technical, policy, and legal fields regarding illicit use of these vehicles beyond their programmed functions. In this paper we explore possible misuse of unmanned vehicles and illustrate the need for research in the technical, policy, and legal realms.

**Date** 13-14 Nov. 2018

**DOI** 10.1109/ISTAS.2018.8638276

**Extra** Journal Abbreviation: 2018 IEEE International Symposium on Technology and Society (ISTAS)

**ISBN** 2158-3412

**Pages** 60-65

**Proceedings Title** 2018 IEEE International Symposium on Technology and Society (ISTAS)

**Conference Name** 2018 IEEE International Symposium on Technology and Society (ISTAS)

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### Tags:

Autonomous vehicles, Law enforcement, Privacy, Unmanned aerial vehicles, Cyber Security, Cyber Law, Cyber Policy, Unmanned Vehicles

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## Trust and Technology Acceptance in Crossing-Decision of Pedestrian in Automated Vehicles

**Item Type** Conference Paper

**Author** S. Pal

**Author** L. Sharma

**Author** M. Jangid

**Abstract** When crossing the street, maximum pedestrians initiate eye contact with the driver when they perceive an oncoming car. This is one of the traditional methods of ensuring that the driver has noticed you. It is critical to be visible to the driver to cross the street safely and maintain trust and acceptance between pedestrians and fully autonomous vehicles. Assume the same circumstance, except instead of making eye contact with the driver, you realise that the vehicle is without one of the drivers. Would you ever prefer to cross the road without establishing a foundation of trust and acceptability of new technology to feel a sense of security? This paper focuses on and addresses the role of trust and technology acceptability in influencing road users' crossing decisions to offer a sense of security.

**Date** 25-26 March 2022

**DOI** 10.1109/ICACCS54159.2022.9785157

**Extra** Journal Abbreviation: 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)

**Volume** 1

**ISBN** 2575-7288

**Pages** 609-614

**Proceedings Title** 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)

**Conference Name** 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)

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## Tags:

Autonomous vehicles, Autonomous Vehicles, Trust, Security, Vehicles, Automobiles, Technology Acceptance, Pedestrian Security, Communication systems, Roads

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## Unveiling Topics from Scientific Literature on the Subject of Self-driving Cars using Latent Dirichlet Allocation

**Item Type** Conference Paper

**Author** W. Y. Ayele

**Author** G. Juell-Skielse

**Abstract** Self-driving cars are becoming popular topics in academia. Consumers of self-driving cars and vehicles have different concerns, for example, safety and security, to name a few. Also, the public sector has interests in self-driving cars such as amending policies to enable the management of self-driving vehicles in cities, urban planning, traffic management and, etc. In this paper, more than 2700 corpus are extracted from literature from several subject areas to identify latent (hidden) topics of self-driving cars. Latent Dirichlet Allocation (LDA) is used for topic identification. The result of this study shows that topics identified are valid research areas such as urban planning, driver car (computer) interaction, self-driving control and system design, ethics in self-driving cars, safety and risk assessment, training dataset quality and machine learning in self-driving cars are among the topics identified. Furthermore, the network visualization of association

graph of terms shows that the most frequently discussed concepts reveal that control of self-driving cars is based on algorithms, data, design, method, and model. The methods used in this study and the results can be used as decision tools, if carefully applied, in diverse disciplines that are disrupted by the introduction of self-driving cars. For future study, we plan to extend this study with a larger dataset and other data mining techniques.

**Date** 1-3 Nov. 2018

**DOI** 10.1109/IEMCON.2018.8615056

**Extra** Journal Abbreviation: 2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)

**Pages** 1113-1119

**Proceedings Title** 2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)

**Conference Name** 2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)

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**Modified** 20/09/2026, 7:46:33 pm

### Tags:

Safety, Automobiles, self-driving cars, Machine learning, topic modeling, Autonomous automobiles, Resource management, unsupervised topic modeling, unveiling hidden topics, Vocabulary

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## User-Centric Security and Privacy Threats in Connected Vehicles: A Threat Modeling Analysis Using STRIDE and LINDDUN

**Item Type** Conference Paper

**Author** B. Stingelová

**Author** C. T. Thrakl

**Author** L. Wrońska

**Author** S. Jedrej-Szymankiewicz

**Author** S. Khan

**Author** D. Svetinovic

**Abstract** The increasing equipment of cars with smart systems and their networking with other devices is leading to a growing network of connected vehicles. Connected cars are Internet of Things (IoT) devices that communicate bidirectionally with other systems, enabling internet access and data exchange. Artificial Intelligence (AI) offers benefits such as autonomous driving, driver assistance programs, and monitoring. The increasing connectivity of cars also brings new risks to users' privacy. Our study focuses on privacy threats in connected cars from a user perspective. Our study provides a comprehensive threat model analysis based on a combination of STRIDE and LINDDUN. We analyze the various threats and vulnerabilities that arise from connecting cars to the internet and other devices, including Vehicle-to-Vehicle (V2V), Vehicle-to-Vlound (V2C), and Vehicle-to-Device (V2D). We conduct our study based on a theoretical model of a modern-day connected vehicle of another study. Our study shows that several types of threats can negatively impact the privacy of connected car users. This encapsulates the potential risks, such as the inadvertent disclosure of personal

data due to the vehicle's interconnectedness with other devices, including smartphones, and the subsequent susceptibility to unauthorized access, while also highlighting the need for robust security measures indicated by our comprehensive threat modeling, to safeguard against a wide array of identified cybersecurity threats.

**Date** 14-17 Nov. 2023

**DOI** 10.1109/DASC/PiCom/CBDDCom/Cy59711.2023.10361381

**Extra** Journal Abbreviation: 2023 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDDCom/CyberSciTech)

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**Conference Name** 2023 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDDCom/CyberSciTech)

**Date Added** 20/09/2026, 7:46:33 pm

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## Tags:

Automobiles, Vehicular ad hoc networks, Privacy, Object recognition, Artificial Intelligence, IoT, Connected vehicles, Internet of Things, Connected Cars, Threat modeling, Cyberphysical Systems, Threat Modeling

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## Vehicle-Life Interaction in Fog-Enabled Smart Connected and Autonomous Vehicles

**Item Type** Journal Article

**Author** B. Feroz

**Author** A. Mehmood

**Author** H. Maryam

**Author** S. Zeadally

**Author** C. Maple

**Author** M. A. Shah

**Abstract** Traffic accidents have become a major issue for researchers, academia, government and vehicle manufacturers over the last few years. Many accidents and emergency situations frequently occur on the road. Unfortunately, accidents lead to health injuries, destruction of some infrastructure, bad traffic flow, and more importantly these events cause deaths of hundreds of thousands of people due to not getting treatment in time. Thus, we need to develop an efficient and smart emergency system to ensure the timely arrival of an ambulance service to the place of the accident in

order to provide timely medical help to those injured. In addition, we also need to communicate promptly with other entities such as hospitals so that they can make appropriate arrangements and provide timely medical information to emergency personnel on the scene including alerting those related to the injured person(s). In this paper, we have developed an intelligent protocol that uses connected and autonomous vehicles' scenarios in Intelligent Transportation System (ITS) so that prompt emergency services can be provided to reduce the death rate caused. The proposed protocol smartly connects with all the relevant entities during the emergency while maintaining a smooth traffic flow for the arrival of the ambulance service. Moreover, our protocol also mitigates the broadcasting of messages circulating over the network for delay sensitive tasks. The evaluation results, based on the performance metrics such as channel collision, average packet delay, packet loss, and routing-overhead demonstrate that our proposed protocol outperforms previously proposed protocols such as Emergency Message Dissemination for Vehicular (EMDV), Contention Based Broadcasting (CBB), and Particle Swarm Optimization Contention-based Broadcast (PCBB) protocols. Finally, we discuss several issues and challenges that need to be addressed in the network in order to achieve more a reliable, efficient, connected, and autonomous vehicular network.

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**DOI** 10.1109/ACCESS.2020.3049110

**Volume** 9

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### Tags:

Autonomous vehicles, Accidents, intelligent transportation system, Connected vehicles, VANETs, Hospitals, Roads, Accident, ambulance service, Delays, Routing protocols, smart emergency

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## Vision or Threat—Awareness for Dual-Use in the Development of Autonomous Driving

**Item Type** Journal Article

**Author** S. Schwartz

**Author** L. G. Guntrum

**Author** C. Reuter

**Abstract** In the digital age, the vision of autonomous vehicles (AVs) is vibrant. Research is being conducted worldwide to integrate AVs into our everyday lives in the future, spending considerable amounts of money in the development process. Actors from both engineering as well as social sciences are involved in this research, with technical disciplines strongly dominating. In addition to perceived progress of numerous newly developed technologies such as AVs, challenges should also be referred to. According to research analysis, the transferability of autonomous cars to the military sphere seems to be frequently forgotten or ignored (dual-use). Since not

much research has been conducted in Germany on the potential deployment of autonomous driving development steps into military domains, 25 semistructured interviews with developers and researchers and actors involved in the field, were conducted in 2020. This article identifies that the majority of respondents interviewed were aware of general existing dual-use debates; however, few had reflected about dual-use issues regarding a possible transfer of their own development processes in the context of autonomous driving to military applications, intensively. One reason is the small-scale nature of research, another is the complexity of the field, which enables the engineer's alienation from their responsibility for the artefacts' use. Moreover, it has become clear that hardly any conversations among colleagues occur about possible misuse and that no standardized policy guidelines exist, which provide information about possible risk. To raise dual-use awareness, scientific contributions, risk education, and interdisciplinary discussions are essential.

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**DOI** 10.1109/TTS.2022.3182310

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## Tags:

Autonomous vehicles, Autonomous driving, Research and development, Automotive engineering, Industries, Software, Ethics, awareness raising, dual-use, risk education, Weapons

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## Voice Design and Trust in Automated Vehicles: Findings and a Research Agenda

**Item Type** Conference Paper

**Author** J. Chen

**Author** Q. Zhang

**Abstract** Most work on human-automated vehicle (AV) interaction has asked what an AV should explain rather than how it should sound while explaining. Building on this gap, we test how voice gender, user-voice gender similarity, and gender-role congruity shape cognitive and affective trust in AVs. In an online experiment with 326 U.S. drivers who viewed six driving scenarios with “what + why” explanations delivered by synthetic male or female voices, we found that gender similarity increased affective trust but did not change cognitive trust; this affective gain appeared only when the voice fit the role participants perceived the AVs to occupy, and it disappeared under role incongruity. Interpreted through Computers Are Social Actors (CASA), similarity attraction, and role congruity theory, the results identify distinct pathways for trust formation: emotional comfort is sensitive to social alignment and role expectations, whereas competence judgments rely on informational quality. We propose a research

agenda that separates cognitive and affective trust, advances fairness-aware voice design beyond binary gender through non-gendered and androgynous options, and specifies context-adaptive communication that adjust tone, framing, and explanation depth to situational demands while preserving user agency. Methodologically, we recommend using validated instruments to measure both facets of trust, incorporating behavioral metrics that capture calibration and safety relevance, developing reproducible voice datasets with detailed parametric metadata, and conducting pre-registered, multi-site studies. The goal is to develop AV voices that earn trust appropriately by clarifying reasoning when needed, offering social reassurance when helpful, and remaining inclusive across users and contexts.

**Date** 12-14 Nov. 2025

**DOI** 10.1109/TPS-ISA67132.2025.00085

**Extra** Journal Abbreviation: 2025 IEEE 7th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)

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**Proceedings Title** 2025 IEEE 7th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)

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Safety, Security, Vehicles, Intelligent systems, Privacy, automated vehicles, Metadata, Shape, Buildings, Measurement, affective trust, cognitive trust, fairness-aware design, human-AI collaboration, Instruments, stereotype-free interfaces, trust evaluation metrics, voice design